

***IN-SILICO* DESIGNING OF A MULTI-EPITOPE VACCINE AGAINST CYTOMEGALOVIRUS**

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DECLARATION

I, **Vishakha Singh**, declare that this training report, entitled “***In-Silico Designing of a Multi-Epitope Vaccine Against Cytomegalovirus***,” is a genuine piece of my work. This diligent work was brought to a close under the vital supervision and direction of **Dr. AJIT KUMAR**, Associate Professor, Centre for Bioinformatics, Maharshi Dayanand University, for the partial fulfillment of the degree of Bachelor of Technology in Biotechnology, University Institute of Engineering and Technology, Maharshi Dayanand University, Rohtak-124001, Haryana.

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Certificate

This is to certify that the dissertation entitled “***In-silico* Designing of Multi-Epitope Vaccine Against Cytomegalovirus**” being submitted by **Ms. Vishakha Singh** (University Roll No. **2078068** to the **University Institute of Engineering and Technology, Maharshi Dayanand University, Rohtak** for the partial fulfilment of the requirement of the degree of Bachelor of Technology in Biotechnology. This is a bonafide work carried out by her at Toxicology and Computational Biology Group, Centre for Bioinformatics, Maharshi Dayanand University, Rohtak. The matter embodied in this industrial training report has not been submitted elsewhere in part or full for the award of any other degree or diploma.

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CHAPTER 1- INTRODUCTION

Human Cytomegalovirus (HCMV), or *Human Herpesvirus 5*, is a member of the *Herpesviridae* family. It is characterized by a large double-stranded DNA genome (~235 kb) encoding over 200 proteins (Davison *et al.*, 2003; Dolan *et al.*, 2004). Following primary infection, HCMV establishes lifelong latency, primarily in monocytes and myeloid progenitor cells, with potential for reactivation, especially in immunocompromised hosts.

Although often asymptomatic in healthy individuals, HCMV poses a serious threat to immunocompromised patients and is the leading cause of congenital infections globally, contributing to long-term neurological disabilities such as sensorineural hearing loss and intellectual impairment (Kenneson & Cannon, 2007).

HCMV is one of the major global health threats, and it is present in a significant portion of the world's population. The results of seroprevalence studies show that an average of 50% to 90% of the global adult population is infected with HCMV over time. The variations in the regions become apparent and are affected by age, socioeconomic status, hygiene practices, and geographic locations. The studies demonstrate that the prevalence rates are clearly higher in low- and middle-income countries, especially reaching 90% or even more, and lower, that is within 40–60% frequency in high-income regions, as shown in Table 1.1. For example, HCMV is a virus (if you narrow it down) that contributes the most to a major section of diseases caused by infectious agents in fetuses and is responsible for the occurrence of 0.5% to 2% of every 200 live births. It should be taken into account that some of these newborns will be symptomatic for CMV-related congenital defects, a characteristic of less than 15% of confirmed cases, while half of the initially asymptomatic cases may experience late sequelae such as sensorineural hearing loss, neurodevelopmental delays, or vision impairment. Besides, HCMV reactivation persists as a critical issue in the immunocompromised, where it is one of the main reasons for their suffering and death. Especially, retinitis, pneumonitis, and gastrointestinal disease might become the disorders that poison a person's life if we talk about solid organ transplant and hematopoietic stem cell transplant recipients, or those suffering from HIV/AIDS. Human Immunodeficiency virus (HIV) serves as a catalyst for reactivation during its active phase. The emergence and the uniqueness of the disease, which are constantly changing, at least in part, due to the high prevalence of the disease characteristics and the dynamics of the disease, make it have a strong need for an effective and accessible vaccine. There is a dire need for an effective and available vaccine, given the wide range of people's exposure to HCMV and the disability it can cause to their immune systems (Zuhair., 2019).

The virus employs sophisticated immune evasion strategies, including downregulating MHC molecules and modulation of cytokine responses, making vaccine development particularly challenging (Jackson *et al.*, 2011). Despite intensive efforts, no licensed vaccine is currently available, underscoring the need for novel approaches like in silico and epitope-based vaccine design (Plotkin *et al.*, 2014).

Despite extensive research, no licensed HCMV vaccine exists to date. Conventional strategies—including live-attenuated, subunit, and vector-based vaccines—face major limitations. Live-attenuated vaccines pose safety risks in immunocompromised individuals, while subunit vaccines often lack broad and lasting immunity due to HCMV's complex antigen profile and immune evasion. The virus encodes proteins that suppress MHC I/II expression, inhibit NK cell responses, and disrupt cytokine signaling, undermining vaccine efficacy (Plotkin *et al.*, 2014; Jackson *et al.*, 2011).

The complexity of HCMV and limitations of conventional vaccine strategies have led to the emergence of computational vaccine design as a promising alternative. In silico approaches such as reverse vaccinology and bioinformatics tools enable rapid identification of conserved, antigenic, and immunodominant epitopes from complete proteomes (Rappuoli *et al.*, 2000). These methods also allow evaluation of population coverage, antigenicity, allergenicity, and structural modelling of vaccine constructs without the need for pathogen culture or expensive animal models. This approach is particularly suited to viruses like HCMV, where immune evasion and latency make empirical vaccine development extremely challenging.

The purpose of this study is to develop a multi-epitope vaccine candidate against HCMV using in silico tools, focusing on the identification of conserved, immunogenic epitopes and computational validation of vaccine efficacy and safety.

Table 1.1 - HCMV Seroprevalence by Region

<i>S.NO.</i>	<i>Region</i>	<i>Estimated Seroprevalence (%)</i>	<i>Notes</i>
1.	Africa	90–100%	High prevalence is attributed to factors like socioeconomic conditions and healthcare access.
2.	Asia	90–100%	Elevated rates have been observed in countries like China and Sri Lanka.
3.	Eastern Mediterranean	90% (95% UI: 85–94)	Includes countries in the Middle East and North Africa.
4.	Western Pacific	80–90%	Encompasses regions like Southeast Asia and Oceania.
5.	Americas	60–80%	Variability exists between North and South America.
6.	Europe	66% (95% UI: 56–74)	Lower prevalence compared to other regions, possibly due to better healthcare infrastructure
7.	United States	50–80%	Seroprevalence increases with age; about 50% by age 40.
8.	Sri Lanka	94.9%	High prevalence among various population groups.
9.	Brazil	81–90%	Studies indicate high seroprevalence in urban populations.

CHAPTER 2- REVIEW OF LITERATURE

2.1 Taxonomic Classification and Genomic Organization of HCMV

HCMV is referred to as *Human Herpesvirus 5 (HHV-5)* and is part of the *Betaherpesvirinae* subfamily within the *Herpesviridae* family. Similar to other betaherpesviruses, HCMV illustrates a prolonged replication cycle, a wide cellular tropism, and the capability to induce lifelong latency in the host, mainly in myeloid progenitor cells (Mocarski *et al.*, 2013). It is highly adapted to a certain species and is the most genetically intricate virus that has been found to infect humans.

HCMV's genome is a linear double-stranded helical DNA molecule of around 235 kilobase pairs(kbp) that presents above 200 reading frames (Dolan *et al.*, 2004). The majority of these frames encode immunomodulatory or host-interactive proteins. The arrangement of the genome is divided into two types — UL (unique long) and US (unique short) — and at the ends of both, there are inverted and direct repeats that support the process of genomic isomerization, leading to high variability of the sequence. As it is necessary for their replication (e.g., UL54), their immune escape (e.g., UL18, UL40), and their latency (e.g. UL138) by which they are known, are still the most popular as they are thus still of significant relevance in the understanding of the pathogenicity of the virus.

It is not just that the HCMV genome only encodes proteins, but also encodes some non-coding RNAs, even a class of small nucleolar RNAs (snoRNAs), such as viral microRNAs. Since these small RNAs modulate the functions of the host cell that are damaged by the virus and also change the innate immune system of the host, the outcome of these changes is an increased level of the virus expulsion, as has been proven in experiments that include both animal models and human cells in vitro as per (Gatherer *et al.*, 2011). Due to the mutations, recombination, and adaptation to the different hosts, traditional vaccine generation, which is based on a stationary genome, always fails to resist the constant changes of naturally occurring variants, most of which are actually under the selective pressure of the immune system and the drugs prescribed.

HCMV exists as several strains, with distinct differences in pathogenicity and genome structure. The most widely studied laboratory strains include:

- **AD169:** A highly passaged strain, often used in research due to its well-characterized genome. It is known for its reduced virulence in vivo and is frequently used in vaccine development studies. However, its high passage history has led to genetic modifications that may not represent natural isolates (Dolan *et al.*, 2004; Mocarski *et al.*, 2013).

- **Towne:** Another laboratory strain, developed from AD169, and known for its attenuated nature, making it a candidate for vaccine development. The Towne strain is commonly used in clinical vaccine trials due to its relatively mild pathogenicity (Dolan *et al.*, 2004).
- **Merlin:** A clinical strain that is less passaged than AD169 and Towne, providing a more representative model of natural HCMV infection. The Merlin strain is used to study the virus's interactions with the host immune system and its ability to evade immune responses (Davison *et al.*, 2003).

In addition to these laboratory strains, clinical isolates of HCMV vary significantly in their genetic makeup. Genomic sequencing of clinical strains has revealed a diverse pool of HCMV variants, with differences in the presence of certain genes, such as those involved in immune evasion and host adaptation. These variations are crucial for understanding the virus's behaviour in different populations and its potential to evade immunity.

2.2 Structural Architecture of Human Cytomegalovirus

Human cytomegalovirus (HCMV) is a large, enveloped virus with a complex architecture typical of the *Herpesviridae* family. The virion measures approximately 200–300 nm in diameter and consists of four concentric layers: a central core containing the linear double-stranded DNA genome, an icosahedral capsid, a surrounding proteinaceous tegument, and an outer lipid envelope embedded with glycoproteins. This multilayered structure is essential not only for viral stability but also for host cell entry, immune evasion, and replication (Mocarski *et al.*, 2013).

2.2.1 Capsid and Core

At the center of the HCMV virion lies a dense, linear double-stranded DNA genome approximately 235 kilobase pairs (kbp) in size. This genome is enclosed within an icosahedral capsid with a diameter of ~100 nm. The capsid is composed of 162 capsomeres arranged in T=16 symmetry, and its assembly is primarily orchestrated by the major capsid protein (MCP; encoded by UL86), along with other structural components such as the smallest capsid protein (SCP; UL48.5) and the triplex proteins (UL46 and UL85). The capsid provides mechanical protection to the viral genome during transport and infection and facilitates its delivery into the host nucleus upon entry (Yu *et al.*, 2011; Gibson *et al.*, 2008).

2.2.2 Tegument Layer

Surrounding the capsid is the tegument—a unique feature of herpesviruses comprising a thick, asymmetric layer of viral proteins and host factors. This layer serves as a biochemical bridge between the capsid and the envelope and is vital for initiating the early phase of infection by delivering regulatory proteins into the host cytoplasm.

Key tegument proteins include:

- **pp65 (UL83):** The most abundant tegument protein and a dominant antigen recognized by cytotoxic T lymphocytes. It modulates innate immune responses by interfering with antigen processing and presentation, thereby delaying immune detection (Kenneson & Cannon *et al.*, 2007; Sylwester *et al.*, 2005).
- **pp150 (UL32):** A large, essential structural protein that stabilizes the capsid within the tegument. It plays a pivotal role in virion maturation and genome packaging (Marschall *et al.*, 2017).
- **pp71 (UL82):** A transactivator protein that enters the host nucleus immediately after infection. It activates viral immediate-early gene expression by degrading the host cell suppressor Daxx, thus initiating the viral replication cycle (Kim *et al.*, 2016; Kalejta *et al.*, 2008).

The multifunctional nature of tegument proteins—serving structural, regulatory, and immunomodulatory roles—makes them attractive for T-cell-targeted vaccine design, particularly as antigens in vector-based or multi-epitope platforms.

2.2.3 Envelope and Surface Glycoproteins

The outermost layer of HCMV is a lipid bilayer envelope derived from the host Golgi or endosomal membranes, enriched with numerous virally encoded glycoprotein complexes. These glycoproteins mediate virus entry, host range determination, and immune evasion, making them key targets for vaccine development.

The most relevant glycoproteins include:

- **Glycoprotein B (gB; UL55):** A fusogenic envelope glycoprotein essential for viral entry into host cells via membrane fusion. gB is one of the most immunogenic HCMV proteins and the focus of several subunit vaccines, including those combined with MF59 adjuvant in clinical trials (Pass *et al.*, 2009; Baraniak *et al.*, 2018).
- **Glycoprotein H (gH; UL75) and Glycoprotein L (gL; UL115):** These form essential entry complexes in two configurations:
 - **gH/gL/gO (UL74):** Required for viral entry into fibroblasts.

- **gH/gL/UL128–131A (pentameric complex):** Critical for entry into epithelial, endothelial, and myeloid cells. The pentameric complex elicits potent neutralizing antibodies and is considered indispensable for protective vaccine responses (Ryckman *et al.*, 2008; Zhou *et al.*, 2013).
- **Glycoprotein M (gM; UL100):** Identified as a conserved and immunogenic structural protein in the pan-genome analysis (Biswas *et al.*, 2024), gM forms a heterodimer with gN and is involved in virion morphogenesis, envelopment, and immune evasion. Its high antigenicity and solubility led to its selection as one of the four main vaccine targets in their multi-epitope vaccine construct.
- **Glycoprotein N (gN; UL73):** Pairs with gM and plays a role in cell tropism and antigen presentation modulation.
- **Envelope Protein US28:** Another critical structural protein highlighted (Biswas *et al.*, 2024), US28 is a viral G protein-coupled receptor that mimics host chemokine receptors, facilitating immune modulation and aiding immune evasion. Its structural stability, antigenicity, and ability to elicit a humoral response underscore its inclusion in epitope-based vaccine design.

These glycoproteins collectively determine cell entry mechanisms, tissue tropism, and immune system interactions. Their immunodominant nature, surface accessibility, and conservation across strains support their prioritization in recombinant subunit and multi-epitope vaccine strategies.

A total of 54 proteins are encoded in the HCMV genome, and at least 25 membrane glycoproteins are detected on the envelope (Vanarsdall *et al.*, 2012). Infection with HCMV requires a multifaceted entry mechanism between viral glycoproteins and host receptors, which rely on the cell types involved in the tropism of viruses. For viral entry, the HCMV glycoproteins gB and gH/gL are used (Vanarsdall *et al.*, 2012; Stein *et al.*, 2019; Gerna *et al.*, 2019). The proteins UL128, UL130, and UL131 assemble on gH/gL to form the pentameric complex gH/gL/UL128-131, which is responsible for virus infection of epithelial and endothelial cells. The HCMV glycoprotein gO can aid the intracellular transport of gH/gL to the assembly site of the virus but will dissociate from them before virion maturation. The analysis of HCMV gO-mutants confirmed the proposed model in which gH/gL/UL128-131 was responsible for the entry of epithelial/endothelial cells, while the trimeric complex gH/gL/gO was responsible for the entry of fibroblasts. If HCMV glycoproteins gO are missing,

gH/gL of virions will be highly decreased, and so, HCMV cannot enter either fibro (Vanarsdall *et al.*, 2012; Stein *et al.*, 2019; Gerna *et al.*, 2019).

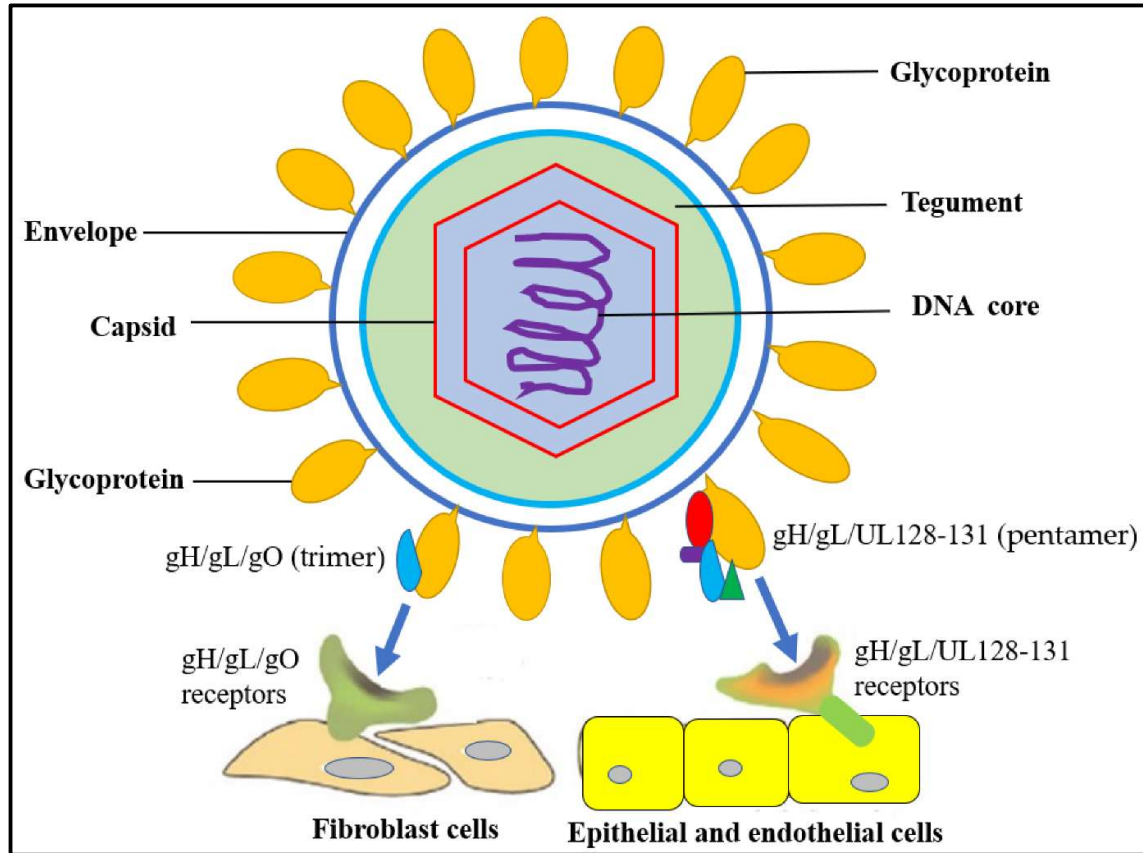


Figure 1.1: Structure of HCMV and entry into host cells. HCMV glycoproteins gH, gL, UL128, UL130 and UL131 assemble to form a functional complex for binding to host cellular receptors, such as gH/gL/UL128- 131 receptors and gH/gL/gO receptors. gH/gL/UL128-131 mediates entry into epithelial/endothelial cells and gH/gL/gO mediates entry into fibroblasts. (Chen *et al.*,2022)

2.3 Clinical Manifestations and Epidemiological Burden

HCMV, a *Betaherpesvirinae* subfamily member of the *Herpesviridae* family, is an omnipresent pathogen having the skill of remain dormant in the human body forever. Although initial HCMV infections usually cause no symptoms in people with a strong immune system, the virus can cause severe diseases like neonatal infections and life-threatening infections in patients with AIDS. The latter case can lead to both more morbidity and healthcare spending on a global scale from HCMV. With both its high prevalence and various forms of spreading, HCMV is

viewed as a constant public health threat. The most susceptible groups in this case are pregnant women, patients with transplants, and persons with AIDS.

2.3.1 Perinatal HCMV Infection and Onset of Pathologies

The congenital cytomegalovirus (cCMV) infection instantly hits home as it represents the top spot in the list of most common congenital infections globally. The disease is the major cause of hearing disabilities in non-genetic children, and that figure is global. In general, the rate at which instances of cCMV happen globally comes to around 0.2–2.0%, but different regions of the world report different levels of it, which can sometimes be very high (Kenneson & Cannon, 2007).

The first two trimesters of the mother being infected are the times with the highest risk of the disease being transmitted to the child and fetal damage occurring as a result. However, in cases when the infection is not primary, such as virus reactivation or reinfection by a different strain, the outcome can be the same, namely, the child will be born with the infection.

Symptomatic infants are suffering from a group of clinical signs such as petechiae, hepatosplenomegaly, jaundice, microcephaly, ventriculomegaly, and intrauterine growth restriction (Dollard *et al.*, 2007). Neurological abnormalities, calcifications in periventricular regions of the brain, and ocular defects are also common. Long-term sequelae, e.g., cognitive impairment, cerebral palsy, and progressive SNHL, occur in 40–60% of symptomatic infants, and in about 10–15% of the asymptomatic at birth, respectively (Manicklal *et al.*, 2013). These adverse effects induce severe functional disability and impose lifelong demands on the healthcare and educational systems.

2.3.2 HCMV in Immunocompromised Populations

Persons who have a weakened immune system, in particular the patients that have received treatments like hematopoietic stem cell transplantation (HSCT), solid organ transplantation (SOT), or people infected by HIV/AIDS, are in a high-risk group of patients who might develop HCMV disease due to the reactivation of the virus. In the case of transplanted patients, HCMV has been associated with various complications such as viral syndrome, tissue-invasive disease (such as CMV colitis, pneumonitis, and hepatitis), graft dysfunction, and increased susceptibility to secondary infections (Kotton *et al.*, 2013). Moreover, post-transplant CMV disease is related to excessive hospitalizations, mortality, and reduced graft and patient survival rates.

CMV retinitis was one of the most frequent opportunistic infections in patients with advanced HIV/AIDS before the availability of highly active antiretroviral therapy (HAART). If not

treated, retinitis can be the source of progressive vision loss and trigger permanent blindness (Jacobson *et al.*, 2000). Although antiretroviral therapy led to a significant decrease in CMV-related opportunistic infections, the virus persistently lingers in latent reservoirs and re-occurs as a risk for patients with immunologic failure.

2.3.3 Global Seroprevalence and Epidemiological Trends

HCMV is a worldwide occurring virus with its seroprevalence being influenced by factors like mode of living, economic status, customs and habits, and demographic traits. In Western countries such as the USA and most of the states in Europe, the middle-aged population is usually 50-70% seropositive, but in the third world, more than 90% of the people carry antibodies (Cannon *et al.*, 2010). This statistical difference occurs as a result of better hygiene, lower density, earlier maternal age, and breastfeeding. Be that as it may, in a family, the most likely route for transmitting the virus is from one person to another (e.g., from children to their parents) who come into close contact, especially to the one who is attending daycare or preschool. Here, in saliva and urine, infected/affected people can give the virus to others.

Transmission occurs by various means: through blood from mother to child (vertical or congenital), during the time of birth (cervical secretions or breast milk), through saliva, urine, sexual intercourse (horizontal), and from donor to recipient (iatrogenic) (Fishman *et al.*, 2007). The high prevalence and sustained transmission are due to the infectious disease's capacity to stay and be reactivated at regular intervals.

In a meta-analysis that was conducted by Kenneson and Cannon in 2007, it was found that the HCMV seroprevalence the older you get and is a social determinant that is positively related to a country's SES (socioeconomic status). When people reside in crowded places, they can become exposed to the virus that causes the diseases; therefore, the virus, in this case, HCMV, is part of the group that causes the diseases that can be widespread and making the diseases spread widely, hence making it a social determinant of health. By way of example, we can see a high rate of seroprevalence among Hispanic and African American populations in the United States, which is consistent with the extremely high percentage of people from these ethnic groups.

2.3.4 Long-Term Sequelae and Economic Burden.

The clinical and social consequences of HCMV infection extend beyond the immediate perinatal or immunosuppressive period. Children affected by cCMV may require long-term follow-up care, including audiologic surveillance, speech therapy, special education, and neurodevelopmental support services. These interventions incur substantial costs over the

child's lifetime. It was estimated that 25% of children with symptomatic cCMV at birth experience severe developmental disability, often requiring lifelong care (Schleiss *et al.*, 2018). Economically, cCMV contributes to substantial direct and indirect costs. In the United States alone, the estimated annual societal burden of congenital CMV—including healthcare, education, and productivity losses—exceeds \$1.86 billion (Dollard *et al.*, 2007). Similar financial pressures are observed in transplant care, where pre-emptive monitoring, antiviral prophylaxis (e.g., ganciclovir, valganciclovir), and management of CMV disease add significantly to the overall cost of transplantation procedures (Emery *et al.*, 2001).

Nevertheless, the problem of public knowledge of CMV is still not as widespread as with other congenital infections, for example, rubella or toxoplasmosis. Furthermore, the lack of a licensed vaccine or generally effective prophylaxis makes the necessity of strong effort and innovation to be at the centre of HCMV prevention and control strategies even more imperative.

2.4 Historical and Current Vaccine Development Approaches

Despite decades of research, developing an effective vaccine against HCMV remains an unmet global health need. HCMV is a complex betaherpesvirus capable of establishing lifelong latency, and it poses significant health risks to neonates and immunocompromised individuals. The ideal vaccine must elicit robust humoral and cellular immunity against primary infection and reactivation, a challenge exacerbated by the virus's large genome (~235 kb), immune evasion strategies, and strain variability (Takashita *et al.*, 2015).

2.4.1 Live-Attenuated Vaccines

Live-attenuated vaccines were among the first candidates explored for HCMV prevention. The Towne strain, attenuated through serial passage in fibroblasts, showed safety and modest immunogenicity in seronegative adults. However, it failed to protect against wild-type HCMV challenge or prevent congenital infection in clinical trials, likely due to the loss of critical epithelial cell tropism and immune-modulatory genes (Plotkin *et al.*, 1999; Adler *et al.*, 1995). The AD169 strain similarly underwent deletions in the UL/b' region, reducing its relevance as a vaccine candidate (Wilkinson *et al.*, 2008). While these vaccines induced neutralizing antibodies and CD8⁺ T-cell responses, they did not confer sterilizing immunity.

Attempts to improve immunogenicity led to chimeric constructs, such as the Towne-Toledo recombinant viruses, designed to restore missing genomic regions from low-passage strains. However, despite enhanced epitope presentation, their protective efficacy in humans remained

limited (Wang *et al.*, 2004). Moreover, concerns regarding potential reversion to virulence and safety in immunocompromised individuals restricted further clinical deployment.

2.4.2 Subunit Vaccines

Subunit vaccines targeting key surface glycoproteins, especially glycoprotein B (gB), have advanced significantly. gB is essential for viral entry into fibroblasts and a major target of neutralizing antibodies. The most extensively studied gB vaccine, adjuvanted with MF59, demonstrated ~50% efficacy in preventing primary infection in seronegative women (Pass *et al.*, 2009). However, its limitations included rapid waning of antibodies and limited efficacy against reinfection with different strains (Baraniak *et al.*, 2019).

More recently, the HCMV pentameric complex (gH/gL/UL128-131A) has gained attention due to its crucial role in infecting epithelial and endothelial cells, pathways especially relevant in congenital transmission. Inclusion of this complex significantly enhances neutralizing responses in vitro and in vivo (Nelson *et al.*, 2018; Chiuppesi *et al.*, 2017). Formulations that combine gB with the pentameric complex antigens have demonstrated synergistic immunogenicity and are under further investigation.

2.4.3 Viral Vector-Based and Recombinant Platforms

Recombinant viral vectors, such as modified vaccinia Ankara (MVA), adenovirus, and canarypox, have been engineered to express HCMV antigens like gB, pp65, and IE1. These vectors stimulate strong T-cell responses, essential for controlling latent and reactivating infections, particularly in transplant recipients (Wussow *et al.*, 2017; Hansen *et al.*, 2013). Notably, pp65 and IE1 are dominant targets of CD8⁺ and CD4⁺ T cells, respectively, in naturally infected individuals (Sylwester *et al.*, 2005).

Recent advances in nucleic acid vaccine platforms have brought renewed optimism. Moderna's mRNA-1647 vaccine combines six mRNAs encoding full-length gB and the pentameric complex, leveraging the success of lipid nanoparticle-based delivery systems. Phase 1 and 2 trials reported robust neutralizing antibody and polyfunctional T-cell responses, with the vaccine now undergoing Phase 3 trials (Moderna, 2023; Smith *et al.*, 2022).

DNA vaccines such as ASP0113, a bivalent construct encoding gB and pp65, also showed promise in eliciting cell-mediated immunity in transplant patients, although a Phase 3 trial failed to demonstrate a significant reduction in viremia or disease (Griffiths *et al.*, 2017).

2.4.4 Challenges in Translational Success

Despite encouraging preclinical data, several bottlenecks hinder licensure. First, definitive correlates of protection against HCMV infection and disease remain undefined, complicating

efficacy evaluation (Cohen *et al.*, 2019). Second, the high seroprevalence of HCMV complicates the design of placebo-controlled trials, and ethical limitations restrict testing in pregnant women, the primary target group for preventing congenital disease.

Furthermore, antigenic diversity among circulating strains, the presence of immunologically decoy epitopes, and the virus's capacity to downregulate MHC molecules all contribute to suboptimal vaccine efficacy (Hansen *et al.*, 2010). Lastly, variability in vaccine-induced memory responses and the absence of long-lasting mucosal immunity challenge the notion of sterilizing protection, prompting a shift toward vaccines that prevent disease severity rather than infection.

2.5 Emergence of Computational Vaccine Design Approaches

2.5.1 Traditional Vaccine Development and its Limitations

Vaccine development has traditionally relied on empirical strategies such as live-attenuated or inactivated vaccines, subunit protein-based vaccines, and viral vector-based vaccines. While these approaches have proven effective for many diseases, their application to immune-evasive viruses like human cytomegalovirus (HCMV) has been fraught with challenges. Issues such as safety concerns, limited efficacy, and the complexity of pathogen-host interactions have necessitated the exploration of alternative, more precise approaches to vaccine design (Rappuoli *et al.*, 2007; Sette *et al.*, 2010).

2.5.2 Reverse Vaccinology: A Paradigm Shift

Reverse vaccinology is one of the most transformative approaches in the field of vaccine development. This method, which originated in bacterial vaccinology, focuses on identifying potential antigens directly from the genome of a pathogen, bypassing the need for culturing the pathogen. By analyzing the entire genome of a virus such as HCMV, researchers can identify conserved, immunogenic epitopes that are more likely to induce a protective immune response. This computational approach allows for the rapid identification of vaccine candidates based on genomic information (Rappuoli *et al.*, 2007).

2.5.3 Immunoinformatics: Optimizing Immune Response Predictions

Immunoinformatics has emerged as an essential tool in vaccine design by predicting how the immune system will respond to specific antigens. This involves the use of computational algorithms to identify T-cell and B-cell epitopes and assess their antigenicity, immunogenicity, and binding affinity to MHC molecules. Immunoinformatics methods can predict which

peptides are likely to stimulate a strong immune response, allowing for the design of vaccines that are both effective and safe (Ekeruche-Makinde *et al.*, 2012; Lundegaard *et al.*, 2008).

2.5.4 Advantages of Computational Approaches

The application of computational vaccine design provides several significant advantages over traditional methods. These advantages include reduced time and cost associated with vaccine development, as well as the ability to identify and design antigens with high specificity and minimal risk of adverse effects. Moreover, computational tools can predict the population coverage of a vaccine by considering genetic diversity and HLA variation, enabling the development of vaccines with global applicability. These *in silico* methods also reduce the need for extensive *in vitro* and *in vivo* experimentation, accelerating the vaccine discovery process.

2.5.5 Application to Immune-Evasive Viruses Like HCMV

For complex viruses such as HCMV, which employ sophisticated immune evasion strategies and exhibit high genetic diversity, computational approaches offer a promising avenue for vaccine development. The ability to design multi-epitope vaccines by combining various T-cell and B-cell epitopes enhances the breadth and specificity of the immune response, which is critical for targeting immune-evasive pathogens like HCMV. By integrating reverse vaccinology and immunoinformatics, researchers can identify a broader range of potential antigens, improving the overall effectiveness of vaccines against these challenging viruses (Al Zamane *et al.*, 2021; Biswas *et al.*, 2024).

2.5.6 Future Prospects: The Integration of AI and Machine Learning

As computational tools continue to evolve, the integration of artificial intelligence (AI) and machine learning in vaccine design holds great promise. These technologies can improve the predictive accuracy of *in silico* models, optimize epitope selection, and simulate immune responses more effectively. The combination of AI with structural vaccinology approaches may enable the design of even more sophisticated and personalized vaccines for immune-evasive pathogens, including HCMV (Al Zamane *et al.*, 2021).

In conclusion, the rise of computational vaccine design represents a significant advancement in the field, providing researchers with powerful tools to accelerate vaccine development, particularly for complex and immune-evasive pathogens like HCMV. By leveraging genomic, bioinformatics, and immunological data, these approaches offer a more efficient, targeted, and systematic pathway to the development of safe and effective vaccines.

2.6 Bioinformatics Tools Used in Vaccine Design

The advent of computational immunology has brought forth a diverse suite of bioinformatics tools that streamline the vaccine design process. These tools assist in various stages of vaccine development, including epitope prediction, antigenicity assessment, allergenicity evaluation, structural modeling, and molecular docking. Below is an overview of widely used and easily verifiable tools in each category:

2.6.1 Epitope Prediction Tools

Epitope prediction is central to designing subunit or multi-epitope vaccines. Tools like IEDB Analysis Resource (Immune Epitope Database) are extensively used to predict MHC class I and II binding epitopes, as well as B-cell epitopes. The IEDB platform incorporates a range of algorithms, including NetMHCpan and NetMHCIIpan, which have demonstrated high accuracy in predicting T-cell epitopes based on HLA-binding affinity (Reynisson *et al.*, 2020).

2.6.2 Antigenicity and Allergenicity Prediction

Once epitopes are selected, it is essential to assess whether they are antigenic (able to provoke an immune response) and non-allergenic. VaxiJen v2.0 is one of the earliest and most cited alignment-free tools for predicting protective antigens based on physicochemical properties (Doytchinova & Flower *et al.*, 2007). For allergenicity, tools such as AllerTOP v2.0 and AllergenFP v1.0 are commonly employed. These servers use machine learning-based approaches to classify proteins based on known allergen databases (Dimitrov *et al.*, 2014).

2.6.3 Toxicity Assessment

Predicting the potential toxicity of peptide vaccine candidates is essential. Tools like ToxinPred help in screening toxic peptides and modifying sequences to improve safety. It uses SVM-based models trained on experimentally validated toxic and non-toxic peptides (Gupta *et al.*, 2013).

2.6.4 Structural Modeling and Validation

To understand the spatial conformation of the vaccine construct, Swiss-Model and ROBETTA are widely used for 3D structure prediction. ROBETTA uses multiple threading alignments and iterative structural assembly simulations (Yang *et al.*, 2015). Model validation tools such as PROCHECK, ERRAT, and Ramachandran plots from the SAVES server are employed to assess the stereochemical quality of the predicted model.

2.6.5 Molecular Docking and Dynamics

Molecular docking predicts the interaction of vaccine constructs with immune receptors such as TLRs or MHC molecules. Tools like ClusPro and HEX are reliable for protein-protein

docking. ClusPro, for instance, is a fast Fourier transform-based docking server validated by CAPRI experiments (Kozakov *et al.*, 2017).

2.6.6 Population Coverage Analysis

Given the polymorphic nature of HLA alleles across different ethnic groups, IEDB's population coverage tool is invaluable in predicting the percentage of individuals in various populations that can respond to the selected epitopes (Vita *et al.*, 2019). This ensures global applicability of the vaccine construct.

2.7 Previous *In-Silico* Vaccine Design Studies on HCMV

The application of *in silico* methods has significantly advanced the field of vaccine development against human cytomegalovirus (HCMV). Due to the virus's ability to establish latency, evade immune surveillance, and exhibit strain-specific genetic diversity, conventional vaccine approaches have faced substantial limitations. Computational vaccine design offers a faster, more targeted alternative by enabling the prediction of epitopes, structural modeling, and immune simulation, all without the need for extensive wet-lab experimentation.

2.7.1 Multi-Epitope Vaccine Design

One of the most promising *in silico* strategies involves the design of multi-epitope vaccines. These constructs incorporate a combination of cytotoxic T lymphocyte (CTL), helper T lymphocyte (HTL), and B-cell epitopes to induce both cellular and humoral immune responses. In a study by Khan *et al.* (2021), researchers identified antigenic regions from several HCMV proteins, including glycoprotein B, pp65 (UL83), and IE1 (UL123), which are known to elicit strong immune responses in natural infections. The selected epitopes were evaluated for antigenicity, allergenicity, and toxicity using tools like VaxiJen, AllerTOP, and ToxinPred.

The final vaccine construct was assembled using appropriate linkers (AAY, GPGPS, and KK) to enhance epitope presentation and processing. A suitable adjuvant, such as the TLR-4 agonist β -defensin, was added to the N-terminal of the construct to boost immunogenicity. Structural modeling was performed using I-TASSER and refined with GalaxyRefine, while docking studies with immune receptors (TLR4 and MHC alleles) demonstrated favorable interactions. The construct also showed high population coverage across diverse HLA haplotypes, suggesting its potential applicability on a global scale.

2.7.2 Pan-Genomic and Conserved Epitope-Based Strategies

A more recent and comprehensive study by Biswas *et al.* (2024) introduced a pan-genome-based reverse vaccinology pipeline to design an HCMV vaccine candidate. This approach

involved analyzing multiple complete genomes of HCMV strains to identify conserved proteins that could serve as universal vaccine targets. The researchers shortlisted proteins that were surface-exposed, essential for virulence, and non-homologous to human proteins—criteria essential to avoid autoimmunity and ensure pathogen specificity.

Subsequent immunoinformatics analyses predicted CTL, HTL, and B-cell epitopes from these conserved proteins. These epitopes were screened for antigenicity, non-toxicity, and solubility. The finalized multi-epitope construct was modelled and validated using tertiary structure prediction tools, and immune simulations were carried out using the C-ImmSim server. The simulated immune response showed a significant rise in IgG and IgM levels, along with increased memory T-cell populations—indicators of long-term protective immunity.

2.7.3 Focus on Immunodominant Targets: pp65 and Glycoprotein B

Multiple *in-silico* studies have centred on pp65 (UL83), a major tegument protein of HCMV, due to its role as a dominant target for cytotoxic T-cell responses. Experimental evidence has confirmed that pp65 elicits strong and consistent responses in HCMV-seropositive individuals, making it a recurrent feature in computational vaccine designs. Likewise, glycoprotein B (gB), involved in viral entry and cell fusion, has been targeted for B-cell epitope prediction, aiming to stimulate neutralizing antibody production.

These proteins have been used either individually or in combination in epitope prediction pipelines, and their inclusion has consistently improved the immunogenic potential of the designed vaccines.

2.7.4 Challenges and the Path Forward

While these studies provide promising theoretical vaccine candidates, none have progressed to preclinical or clinical stages. The key challenges remain the translation of *in silico* predictions into biological efficacy. Factors such as epitope processing, HLA presentation variability, and immunodominance hierarchy require experimental validation.

Nevertheless, the computational models serve as a powerful foundation, significantly reducing the time and cost associated with initial vaccine discovery. By narrowing down candidate epitopes to the most promising targets, *in silico* pipelines provide a rational starting point for laboratory-based testing.

2.8 Future Directions in Computational Vaccine Design

The future of vaccine development is shaped by rapid advancements in computational biology, artificial intelligence (AI), and systems immunology. Computational approaches are expected

to play an increasingly central role as conventional methods encounter limitations, particularly in addressing complex and immune-evasive viruses like human cytomegalovirus (HCMV). AI and machine learning are enhancing the predictive capabilities of bioinformatics tools by enabling accurate identification of T-cell and B-cell epitopes, optimizing antigen selection, and forecasting immune responses with greater precision (Baruah *et al.*, 2023; Guan *et al.*, 2019). Simultaneously, breakthroughs in structural bioinformatics—enabled by tools like AlphaFold2—are facilitating the prediction of conformational epitopes critical for neutralizing antibody responses (Jumper *et al.*, 2021).

Incorporating population-specific HLA data into epitope prediction allows for the design of vaccines with broader population coverage, while personalized vaccine strategies tailored to individual immune profiles are also emerging as viable avenues (Yadav *et al.*, 2022). Moreover, immune simulation platforms are being refined to evaluate vaccine-induced responses *in silico*, helping to reduce dependence on early-stage animal testing (Rapin *et al.*, 2010). The integration of multi-omics data, including genomics, transcriptomics, and proteomics, is also expected to support the discovery of novel and conserved vaccine targets, especially relevant to pathogens with latent phases such as HCMV (Pulendran *et al.*, 2010).

Greater emphasis is required on experimental validation, collaborative data sharing, and the development of standardized *in-silico* pipelines to bridge the gap between computational predictions and clinical application. Collectively, these advancements offer a promising outlook for designing safer, more effective, and globally relevant vaccines against HCMV and similar viral pathogens.

CHAPTER 3: OBJECTIVE

Aim:

This study aims to design a multi-epitope vaccine against Human Cytomegalovirus (HCMV) using *in-silico* techniques to optimize antigen selection and vaccine efficacy.

Objectives:

- Retrieve and analyze the HCMV AD169 strain proteome to identify potential antigenic proteins and prediction of CTL, HTL, and B-cell epitopes.
- Construct a multi-epitope vaccine candidate and assess it's physicochemical properties, antigenicity, allergenicity, and immunogenicity.
- Modelling of the vaccine construct, validate its structure and perform protein-protein docking with immune receptors.

CHAPTER 4: MATERIALS & METHODOLOGY

4.1 MATERIALS

1. **NCBI- The National Centre for Biotechnology Information**

(<https://www.ncbi.nlm.nih.gov/>) is a comprehensive and widely used bioinformatics resource that provides access to genomic, proteomic, and biomedical literature databases. In this study, NCBI's GenBank and RefSeq databases were used to cross-verify the genomic and proteomic data of *Human Cytomegalovirus* (HCMV), specifically the AD169 strain. The corresponding protein accession numbers and sequences obtained from UniProtKB were matched with NCBI entries to ensure annotation accuracy and sequence integrity. NCBI was also utilized to retrieve information on viral gene functions, host interactions, and literature references relevant to the selected proteins, thereby supporting a reliable foundation for downstream epitope prediction and vaccine design.

2. **UNIPROT- The Universal Protein Resource**

(<https://www.uniprot.org/>) is a central hub of protein sequence and functional information, maintained by the UniProt Consortium. This study retrieved the complete proteome of Human Cytomegalovirus (HCMV) strain AD169 from UniProtKB, specifically using the Swiss-Prot section to ensure high-quality, manually curated protein entries. UniProt provides extensive annotations including protein function, domain structure, post-translational modifications, and subcellular localization, all instrumental in the initial screening of candidate antigens. The FASTA-format sequences obtained from UniProt formed the primary dataset for antigenicity prediction and subsequent epitope mapping in the reverse vaccinology pipeline.

3. **Protein Data Bank (PDB)**

(<https://www.rcsb.org/>) is an open-access, international repository of experimentally determined three-dimensional structures of biomolecules, including proteins, nucleic acids, and complexes. In this study, high-resolution crystal structures of immune receptors such as Toll-like Receptor 4 (TLR4) and Major Histocompatibility Complex (MHC) molecules were retrieved from the PDB in .pdb format for use in molecular docking analyses. The selected receptor structures were chosen based on criteria such as completeness, resolution, and biological relevance. These structures served as critical templates for assessing the binding affinity and interaction profile of the designed multi-epitope vaccine construct. The PDB ensured access to validated structural data necessary for accurate docking simulations and visualization of molecular interactions.

4. *VaxiJen v2.0*

(<https://www.ddg-pharmfac.net/vaxijen/VaxiJen/VaxiJen.html>) is an alignment-independent, online server used to predict the antigenic potential of proteins based on their physicochemical properties. It is one of the first tools designed to predict protective antigens without relying on sequence alignment, making it particularly effective for large-scale screening in vaccine design. In this study, VaxiJen v2.0 was employed to assess the antigenicity of HCMV proteins using the “virus” model, with a threshold of 0.5. Proteins with scores above this cutoff were considered probable antigens and selected for downstream epitope prediction. This initial screening helped prioritize proteins likely to elicit a strong immune response, thereby enhancing the efficiency and specificity of the vaccine design process.

5. *AlgPred*

(<https://webs.iitd.edu.in/raghava/algpred/submission.html>) is an integrated bioinformatics tool designed to predict the allergenic potential of proteins and peptides using multiple approaches, including motif-based detection, similarity to known allergens, and machine learning algorithms such as Support Vector Machines (SVM). In this study, AlgPred was used to assess the allergenicity of selected HCMV-derived proteins and predicted epitopes to ensure the safety of the vaccine construct. Only proteins and epitopes classified as non-allergenic by AlgPred were included in the final multi-epitope design. This step was essential to minimize the risk of allergic responses and enhance the safety profile of the proposed vaccine candidate.

6. *ToxinPred*

(<https://webs.iitd.edu.in/raghava/toxinpred/>) is a reliable bioinformatics tool designed to predict the toxicity of peptides and proteins based on their sequence information. Developed by the Raghava Lab, ToxinPred uses machine learning approaches such as Support Vector Machines (SVMs) trained on experimentally validated toxic and non-toxic peptides. In this study, ToxinPred was employed to screen all predicted CTL, HTL, and B-cell epitopes for potential toxicity. Only epitopes classified as non-toxic were selected for inclusion in the final multi-epitope vaccine construct, ensuring the safety and biocompatibility of the designed candidate. This toxicity filtration step was essential to minimize adverse effects and enhance the overall immunological suitability of the vaccine.

7. *IEDB-Immune Epitope Database and Analysis Resource*

(<https://www.iedb.org/>) is a widely used platform that provides tools for the prediction, analysis, and retrieval of experimentally validated immune epitopes for infectious diseases, allergies, and autoimmune disorders. Maintained by the National Institute of Allergy and

Infectious Diseases (NIAID), IEDB integrates a suite of powerful tools that allow prediction of Cytotoxic T Lymphocyte (CTL) epitopes using MHC class I binding algorithms, Helper T Lymphocyte (HTL) epitopes via MHC class II binding predictions, and B-cell epitopes based on linear and conformational determinants. In this study, IEDB was employed to identify potential CTL and HTL epitopes from selected antigenic proteins of HCMV using its recommended consensus and NetMHCIIpan methods, respectively. Additionally, the population coverage tool in IEDB was used to evaluate the global HLA allele representation of the selected epitopes. The use of IEDB ensured that the epitopes included in the multi-epitope vaccine construct were not only immunogenic but also broadly applicable across different human populations.

8. IFNepitope

(<https://webs.iitd.edu.in/raghava/ifnepitope/>) is an online prediction server developed to identify epitopes capable of inducing Interferon-gamma (IFN- γ), a key cytokine in antiviral immunity and Th1-type immune responses. This study used IFNepitope to analyze the shortlisted Helper T Lymphocyte (HTL) epitopes to determine their potential to stimulate IFN- γ production. The server uses a hybrid approach combining motif-based and machine learning (SVM-based) methods for high prediction accuracy. Epitopes predicted as positive IFN- γ inducers were prioritized for inclusion in the vaccine construct to enhance cell-mediated immunity and improve the protective efficacy of the designed multi-epitope vaccine. This step ensured that the vaccine construct not only targets antigen presentation but also boosts a robust immune response through cytokine induction.

9. SWISS-MODEL

(<https://swissmodel.expasy.org/>) is an automated web-based server for homology modeling of protein tertiary structures. It generates high-quality 3D models based on sequence alignment with structurally characterized homologs in the Protein Data Bank (PDB). In this study, SWISS-MODEL was utilized to predict the tertiary structure of the designed multi-epitope vaccine construct. The amino acid sequence of the final construct was submitted in FASTA format, and the server identified suitable templates based on sequence identity and coverage. The generated models were evaluated for structural quality using parameters such as GMQE (Global Model Quality Estimation) and QMEAN scores. The best-scoring model was selected for further refinement and validation. SWISS-MODEL provided a reliable and efficient platform for visualizing and understanding the spatial conformation of the vaccine candidate, essential for downstream docking and simulation analyses.

10. ROBETTA

(<https://robeta.bakerlab.org/>) The Baker Lab developed a powerful and widely used web-based protein structure prediction server at the University of Washington. It integrates both comparative modeling (using known homologous structures) and de novo prediction (for regions without structural templates), making it suitable for modeling novel or chimeric protein sequences such as multi-epitope vaccine constructs. In this study, the final vaccine construct sequence was submitted to Robetta for tertiary structure prediction. The server generated 3D models based on RoseTTAFold, a deep learning-based method for accurate structure prediction. The best model was selected based on confidence scores and structural completeness. Robetta enabled the visualization of the overall folding, domain organization, and surface accessibility of the vaccine, which are crucial for evaluating antigen presentation and receptor binding in subsequent docking and simulation steps.

11. ProtParam

(<https://web.expasy.org/protparam/>) is an online tool provided by the ExPASy server that computes various physicochemical properties of proteins from their amino acid sequences. These properties include molecular weight, theoretical isoelectric point (pI), amino acid composition, instability index, aliphatic index, and grand average of hydropathicity (GRAVY). In this study, ProtParam was used to analyze the multi-epitope vaccine construct to evaluate its stability, solubility, and overall biochemical characteristics. The results helped in assessing the suitability of the vaccine candidate for expression and purification, as well as its potential behaviour in biological systems.

12. MolProbity

(<http://molprobity.biochem.duke.edu/>) is a comprehensive structure validation tool that assesses the quality of protein 3D models by analyzing all-atom contacts, geometry, and steric clashes. It provides detailed feedback on structural features such as Ramachandran plot statistics, rotamer conformations, and bond lengths/angles. In this study, MolProbity was used to evaluate the tertiary structure of the modelled multi-epitope vaccine construct generated by Robetta. The analysis helped identify and rectify potential structural issues, such as unfavourable backbone torsion angles or atomic clashes, thereby refining the model for improved accuracy. This rigorous validation ensured the reliability of the predicted structure for subsequent molecular docking and simulation studies.

13. PROCHECK

[\(https://www.ebi.ac.uk/thornton-srv/software/PROCHECK/\)](https://www.ebi.ac.uk/thornton-srv/software/PROCHECK/) is a widely used bioinformatics tool for assessing the stereochemical quality of protein structures. It analyzes the geometry of the protein backbone and side chains, producing detailed Ramachandran plots to evaluate the distribution of dihedral angles (phi and psi) in the protein model. In this study, PROCHECK was employed to validate the tertiary structure of the designed multi-epitope vaccine construct. The analysis helped identify residues in favoured, allowed, and disallowed regions, providing a quantitative measure of the structural reliability and overall model quality. These validations ensured that the predicted 3D structure is consistent with known protein folding patterns and suitable for further computational analyses.

14. Structure Analysis and Verification Server (SAVE)

[\(https://servicesn.mbi.ucla.edu/SAVES/\)](https://servicesn.mbi.ucla.edu/SAVES/) is an integrated online platform that provides comprehensive validation of protein structures by combining multiple well-established tools, including PROCHECK, ERRAT, VERIFY3D, and WHATCHECK. In this study, the tertiary structure model of the multi-epitope vaccine construct was submitted to the SAVE server for thorough assessment of its stereochemical quality, backbone geometry, and structural compatibility with its amino acid sequence. The combined output from these analyses enabled the identification of structural errors and facilitated refinement steps to improve the accuracy and reliability of the predicted model before downstream applications such as molecular docking and simulation.

15. UCSF Chimera

[\(https://www.cgl.ucsf.edu/chimera/\)](https://www.cgl.ucsf.edu/chimera/) is a highly versatile molecular visualization and analysis software widely used in structural biology. It provides an interactive environment for viewing, analyzing, and manipulating 3D structures of proteins, nucleic acids, and complexes. In this study, UCSF Chimera was used to visualize the tertiary structure of the multi-epitope vaccine construct, as well as the docked complexes with immune receptors. The software facilitated detailed examination of molecular interactions, surface properties, and conformational features. Additionally, Chimera aided in preparing structural files by cleaning, adding hydrogens, and optimizing molecular geometry before docking simulations. Its powerful graphical tools helped generate high-quality images and figures for the presentation and interpretation of structural data.

16. Hex

(<http://hex.loria.fr/>) is an interactive molecular docking software that specializes in protein–protein docking by calculating shape complementarity, electrostatics, and correlation-based scoring using spherical polar Fourier correlations. In this study, Hex 8.0 was employed to evaluate the interaction between the designed multi-epitope vaccine construct and immune receptors such as Toll-like Receptor 4 (TLR4) and Major Histocompatibility Complex (MHC) molecules. The tertiary structure of the vaccine, modeled via Robetta, was docked with the immune receptor structures obtained from the Protein Data Bank (PDB). Docking results were assessed based on binding energy scores and interaction orientations, with lower energy values indicating more stable and favorable interactions. The use of Hex provided valuable insights into the potential immunogenic interactions and receptor recognition of the vaccine candidate.

17. LigPlot⁺

To further analyze and visualize the molecular interactions between the vaccine construct and immune receptors, LigPlot⁺ (<https://www.ebi.ac.uk/thornton-srv/software/LigPlus/>) was used. LigPlot⁺ is a widely used tool for generating two-dimensional (2D) interaction diagrams that illustrate hydrogen bonds and hydrophobic contacts between docked molecules. The docked complex files generated by Hex were processed through LigPlot⁺ to identify key binding residues and interaction patterns at the vaccine–receptor interface. This visualization helped confirm stable binding, highlighting amino acid residues involved in hydrogen bonding and van der Waals interactions, which are critical for effective receptor engagement and immune activation. The use of LigPlot⁺ added clarity to the structural interpretation of docking results and supported the functional relevance of the vaccine–receptor interactions.

4.2 METHODOLOGY

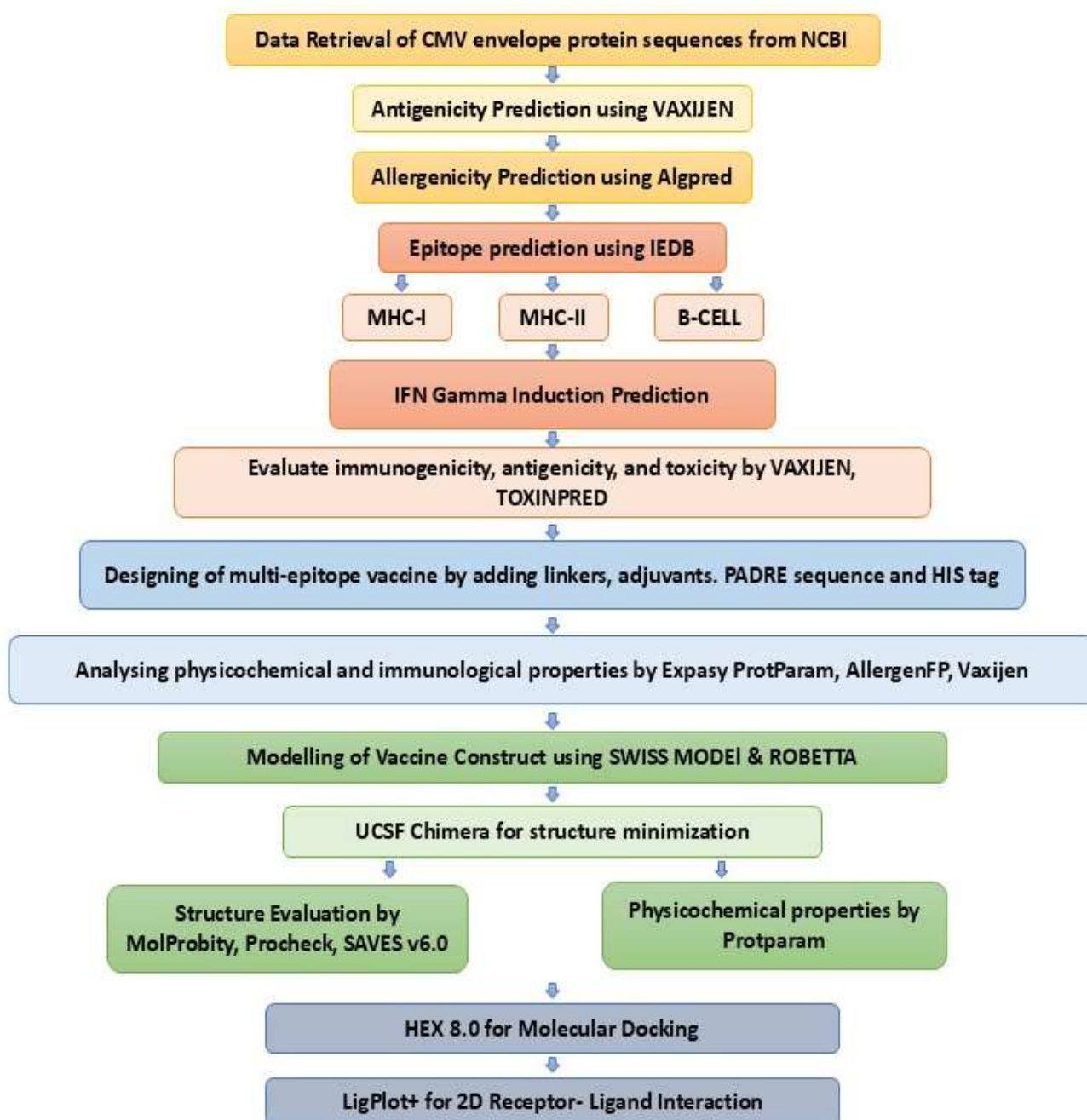


Figure 4.1 Flowchart for the steps of the methodology

CHAPTER 5: RESULTS AND DISCUSSIONS

5.1 Data Retrieval

The protein sequences of *HCMV AD169* was retrieved from the NCBI protein databases with sequence ranges from 311 to 154 amino acids. Proteins sequences retrieved for further studies are listed below:

1. HCMVTRL1 = IRL1 [Human betaherpesvirus 5]
2. HCMVTRL4 = IRL4 [Human betaherpesvirus 5]
3. HCMVTRL14 [Human betaherpesvirus 5]
4. HCMVUL1 [Human betaherpesvirus 5]
5. HCMVUL4 [Human betaherpesvirus 5]

5.2 Antigenicity and allergenicity prediction

All the sequences were evaluated for their antigenic potential using the VaxiJen v2.0 server and the predicted results showed that out of 5 selected proteins 4 had good antigenic property with antigenicity scores of 0.5858, 0.5657, 0.5100 and 0.6542 for HCMVTRL1, HCMVTRL14, HCMVUL1 and HCMVUL4 as mentioned in table 5.1 respectively, which was above the threshold value of 0.4, indicating that the sequence was likely to be antigenic. Additionally, allergenicity assessment was conducted for these proteins using the AlgPred tool indicated that only 2 sequences i.e. HCMVTRL1 and HCMVTRL14 were non-allergenic but as HCMVTRL14 was showing poor antigenicity, therefor only HCMVTRL1 with antigenicity value of 0.5858 supporting its safety profile for potential immunization purposes was selected for further study and multi-epitope construction (as shown in figure 5.1 and 5.2). These findings collectively validate HCMVTRL1 as a strong candidate for further vaccine development.

5.3 Epitope Prediction

CTL and HTL epitopes were predicted using IEDB MHC binding tools. Epitopes demonstrated strong binding affinity ($IC_{50} < 50$ nm) for shortlisted multiple HLA alleles. The B-cell epitopes were predicted based on sequence features and antigenic exposure, ensuring surface accessibility. In B-cell epitope prediction, green regions indicate residues likely to elicit antibody responses. An optimal result shows moderately well-defined green regions, suggesting surface-exposed and accessible epitopes. This balanced distribution enhances immunogenic potential, while too many or too few green areas may indicate non-specificity or

Table 5.1 Predicted allergenicity and antigenicity of the selected protein sequences of AD169 variant of HCMV.

S.No	Sequences	Allergenicity	Antigenicity
1	HCMVTRL1 = IRL1	NON-ALLERGEN	0.5858 (Probable ANTIGEN).
2	HCMVTRL4 = IRL4	NON-ALLERGEN	0.3910 (Probable NON-ANTIGEN).
3	HCMVTRL14	ALLERGEN	0.5657 (Probable ANTIGEN).
4	HCMVUL1	Potential ALLERGEN	0.5100 Probable ANTIGEN).
5	HCMVUL4	ALLERGEN	0.6542 (Probable ANTIGEN).

HCMVTRL1 = IRL1 [Human betaherpesvirus 5]

>CAA35449.1 HCMVTRL1 = IRL1 [Human betaherpesvirus 5]

MPATDTNSTHTTPLHPDAQHTLPLHHSNTQPHVQTSDKHADEEHRTQMELDAADYA
ACAQARQHLYDQTQPLLLAYPNTNPQDSAHPFTENQHQLTHPLHNIGEGAALGYVPV
RAEIRRGGGDWADSASDFDADCWCMWGRFGTMGRQPVVTTLLARQRDGLADWNV
VRCRGTGFRAHDSSEdGVSVWRQHLVFLGGHGRRVQLERPSAGEAQARGLLPRIT
PISTSPRRKPPHPATSTASHHPASPRSDHTLFPVPSTPSATVHNPRNYAVQLHAETTRT
WRWAQRGERGAWMPAETFTCPKDKRPW

Figure 5.1 Sequence of selected HCMV-AD169 protein



Figure 5.2 VaxiJen's predicted antigenicity for HCMVTRL1 [Human betaherpesvirus 5].

low immune-reactivity, as shown in Figure 5.3. Numerous epitopes were predicted from HCMVTRL1 = IRL1 protein using IEDB and categorized as B-cell, MHC-I, and MHC-II. The predicted epitopes were further evaluated and selected on the basis of allergenicity, toxicity, and IFN-Gamma induction score (as mentioned in Tables 5.2 to 5.4). All eight epitopes-comprising B-cell epitopes (B1–B3) and MHC class I and II epitopes (M1–M5), were predicted to be non-allergenic and non-toxic, indicating their safety and suitability for inclusion in the multi-epitope vaccine construct.

Furthermore, IFN- γ inducing potential, a key marker for eliciting a robust Th1-type immune response, was also evaluated. Out of the eight epitopes, six (B2, B3, M3, M4, and M5) showed positive IFN- γ scores, suggesting their ability to stimulate cell-mediated immunity effectively. In contrast, B1 and M1 exhibited negative IFN- γ scores, indicating a lesser likelihood of contributing to IFN- γ production (as mentioned in Table 5.5). Despite this, their inclusion may still provide complementary humoral or helper T-cell responses.

5.4 Multiple-Epitope Vaccine Constructs

A multi-epitope vaccine represents a cutting-edge recombinant strategy that integrates multiple immunogenic components-specifically, cytotoxic T lymphocyte (CTL), helper T lymphocyte (HTL), and B-cell epitopes, into a unified chimeric construct. This approach is designed to elicit a broad and potent immune response by simultaneously stimulating cellular and humoral arms of the immune system. In the present construct, immunogenic epitopes were carefully selected and joined using the RVRR linker, which facilitates proper processing and presentation by MHC molecules. CTL, HTL, and B-cell epitopes were all connected using this linker to preserve their structural integrity and improve accessibility by immune components. An adjuvant was incorporated at the N-terminal of the construct to boost immunogenicity, with options such as L7/L12 ribosomal protein, β -defensin, HBHA, and HBHA conserved domain evaluated for their potential to enhance immune recognition. The adjuvant was fused using the EAAAK linker, known for imparting structural rigidity and functional separation. Additionally, a PADRE sequence (AKFVAAWTLKAAA), known to be a universal T-helper epitope, was included to overcome MHC polymorphism and ensure a consistent T-cell response across diverse populations. This PADRE segment was joined using a GPGPG linker, and all the epitopes were joined with an RVRR linker. To enable straightforward purification, a 6 $\times$ His-tag (HHHHHH) was also appended at the C-terminal of the vaccine construct, allowing affinity-based isolation via nickel chromatography (as mentioned in Table 5.6).

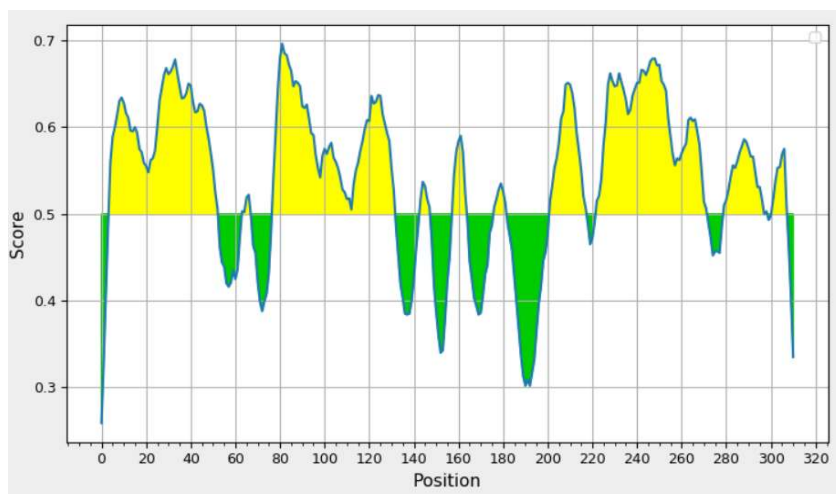


Figure 5.3 Graphical Representation of B-CELL Epitope prediction using IEDB.

Table 5.2 Selected B-CELL Epitopes

Epitope	Sequence	Notes
B1	VQLERPSAGEAQARGLL	Compact & antigenic
B2	FTCPKDK	Minimal + soluble
B3	AHDSER	Acidic + compact

Table 5.3 Selected MHC-I Epitopes

Epitope	Sequence	Description
M-1	HPDAQHTPL	Conserved & antigenic
M-2	EEHRTQMEL	Highly immunogenic
M-3	LPRIRITPI	Strong binding to HLA-I

Table 5.4 Selected MHC-II Epitopes

Epitope	Sequence	Description
M-4	QPLLLAYPNTNPQDS	Broad HLA-DR binding
M-5	HTTPLHPDAQHTLPL	High IFN- γ induction

Table 5.5 Evaluation of predicted epitopes

S.NO.	EPITOPES	ALLERGEN	IFN-GAMMA SCORE	TOXICITY
B1	VQLERPSAGEAQARGLL	Non-allergen	-	Non-Toxic
B2	FTCPKDK	Non-allergen	+	Non-Toxic
B3	AHDSED	Non-allergen	+	Non-Toxic
M1	HPDAQHTPL	Non-allergen	-	Non-Toxic
M2	EEHRTQMEL	Non-allergen	+	Non-Toxic
M3	LPRIRITPI	Non-allergen	+	Non-Toxic
M4	QPLLLAYPNTNPQDS	Non-allergen	+	Non-Toxic
M5	HTTPLHPDAQHTLPL	Non-allergen	+	Non-Toxic

Table 5.6 Various assorted vaccine construct combinations.

S.No	Vaccine composition		Sequence
Vc 1	L7/L12 protein linkers+	Ribosomal +epitopes+ PADRE+ His tag	EAAAKMAKLSTDELLDAFKEMTLLLESDFVKKFEETFEVTAAAPVA VAAAGAAPAGAAVEAAEEQSEFDVILEAAGDKKIGVIKVVREIMSGL GLKEAKDLVDGAPKPLLEKVAKEAADEAKAKLEAAGATVTVKEAA AKGPGPGAKFVAAWTLKAAAAGPGPGHPDAQHTPLRVRRREEHRTQ MELRVRRRLPRIRITPIRVRRQPLLLAYPNTNPQDSRVRRHTTPLHPD AQHTLPLRVRRVQLERPSAGEAQARGLLRVRRFTCPKDRVRRRAHD SEDGPGPGAKFVAAWTLKAAAAGPGPGHHHHHH
Vc2	Beta-defension epitopes+ PADRE+ His tag	+ linkers+	EAAAKGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCTRGRKCC RRKKEAAAKGPGPGAKFVAAWTLKAAAAGPGPGHPDAQHTPLRVRR EEHRTQMELRVRRRLPRIRITPIRVRRQPLLLAYPNTNPQDSRVRRHT TPLHPDAQHTLPLRVRRVQLERPSAGEAQARGLLRVRRFTCPKDRV RRRAHDSSEDGPGPGAKFVAAWTLKAAAAGPGPGHHHHHH
Vc3	HBHA+ linkers+ PADRE+ His tag	epitopes+ His tag	EAAAKMAENPNIDDLPAPLLAALGAADLALATVNDLIANLRERAE TRAETRTRVEERRARLTQFQEDLPEQFIELRDKFTTEELRKAEEGY LEAATNRYNELVERGEAALQRLRSQTAFEDASARAEGYVDQAVELT QEALGTVASQTRAVGERAAKLVGIELEAAAKGPGPGAKFVAAWTLK AAAAGPGPGHPDAQHTPLRVRRREEHRTQMELRVRRRLPRIRITPIRV RQPLLLAYPNTNPQDSRVRRHTTPLHPDAQHTLPLRVRRVQLERPS AGEAQARGLLRVRRFTCPKDRVRRRAHDSSEDGPGPGAKFVAAWTLK AAAAGPGPGHHHHHH
Vc4	HBHA epitopes+ PADRE+ His tag	conserved+ linkers+	EAAAKMAENSNIDDIKAPLLAALGAADLALATVNELITNLRERAEET RRSRVEESRARLTQLQEDLPEQTELREKFTAELRKAEEGYLEAAT SELVERGEAALERLSRQSFEEVSARAEGYVDQAVELTQEALGTVAS QVEGRAAKLVGIELEAAAKGPGPGAKFVAAWTLKAAAAGPGPGHPDA QHTPLRVRRREEHRTQMELRVRRRLPRIRITPIRVRRQPLLLAYPNTNP QDSRVRRHTTPLHPDAQHTLPLRVRRVQLERPSAGEAQARGLLRV RFTCPKDRVRRRAHDSSEDGPGPGAKFVAAWTLKAAAAGPGPGHHHHH H

Overall, the designed multi-epitope vaccine aims to be highly antigenic, non-allergenic, non-toxic, and capable of inducing a globally relevant immune response, offering a precise, rapid, and safer alternative to traditional vaccine approaches against complex pathogens like Human Cytomegalovirus (HCMV).

5.5 Structure modelling and evaluation.

Tertiary structures were initially predicted for all vaccine constructs (Vc1, Vc2, Vc3, and Vc4) using the Robetta server. The predicted structures of all constructs were subjected to energy minimization using UCSF Chimera to improve their geometric stability. Among these, Vc4 exhibited a confidence score of 0.49 and a total length of 325 amino acids. The minimized models were then evaluated using comprehensive structural validation tools, including MolProbity and PROCHECK to assess stereochemical integrity. MolProbity analysis of Vc4 revealed a low clash score, favorable rotamer conformations, and an acceptable overall MolProbity score, indicating good geometric accuracy and minimal structural strain (refer to Figures 5.4 to 5.7).

Further structural validation of Vc4 using Ramachandran plot analysis showed that 91.8% of residues were located in the most favored regions, 7.5% in additional allowed regions, and only 0.4% were in disallowed regions (as shown in Figure 5.8). This distribution of backbone dihedral angles suggests that the model possesses strong stereochemical reliability. Based on these structural quality assessments, the Vc4 Model (as in Figure 5.9) was selected as the most promising vaccine construct and used for downstream applications such as molecular docking and immunological simulation studies.

In-Silico Designing of Multi-Epitope vaccine against Cytomegalovirus

Protein Geometry	Poor rotamers	1	0.43%	Goal: <0.3%
	Favored rotamers	229	99.13%	Goal: >98%
	Ramachandran outliers	4	1.32%	Goal: <0.05%
	Ramachandran favored	285	93.75%	Goal: >98%
	Rama distribution Z-score	-1.15 ± 0.46		Goal: abs(Z score) < 2
	Cβ deviations >0.25Å	0	0.00%	Goal: 0
	Bad bonds:	3 / 2374	0.13%	Goal: 0%
	Bad angles:	6 / 3215	0.19%	Goal: <0.1%
Peptide Omegas	Cis Prolines:	1 / 24	4.17%	Expected: ≤1 per chain, or ≤5%
Low-resolution Criteria	CaBLAM outliers	17	5.6%	Goal: <1.0%
	CA Geometry outliers	6	1.99%	Goal: <0.5%
Additional validations	Chiral volume outliers	0/357		

Figure 5.4 MolProbity analysis of Robetta L7/L12

Protein Geometry	Poor rotamers	0	0.00%	Goal: <0.3%
	Favored rotamers	176	100.00%	Goal: >98%
	Ramachandran outliers	8	3.65%	Goal: <0.05%
	Ramachandran favored	196	89.50%	Goal: >98%
	Rama distribution Z-score	-1.90 ± 0.56		Goal: abs(Z score) < 2
	Cβ deviations >0.25Å	0	0.00%	Goal: 0
	Bad bonds:	1 / 1779	0.06%	Goal: 0%
	Bad angles:	9 / 2402	0.37%	Goal: <0.1%
Peptide Omegas	Cis Prolines:	2 / 21	9.52%	Expected: ≤1 per chain, or ≤5%
	Cis nonProlines:	1 / 199	0.50%	Goal: <0.05%
Low-resolution Criteria	CaBLAM outliers	25	11.5%	Goal: <1.0%
	CA Geometry outliers	9	4.15%	Goal: <0.5%
Additional validations	Chiral volume outliers	0/249		

Figure 5.5 MolProbity analysis of Robetta beta defensin

Protein Geometry	Poor rotamers	0	0.00%	Goal: <0.3%
	Favored rotamers	260	99.62%	Goal: >98%
	Ramachandran outliers	3	0.90%	Goal: <0.05%
	Ramachandran favored	325	97.60%	Goal: >98%
	Rama distribution Z-score	2.23 ± 0.42		Goal: abs(Z score) < 2
	Cβ deviations >0.25Å	0	0.00%	Goal: 0
	Bad bonds:	0 / 2671	0.00%	Goal: 0%
	Bad angles:	6 / 3620	0.17%	Goal: <0.1%
Peptide Omegas	Cis Prolines:	0 / 24	0.00%	Expected: ≤1 per chain, or ≤5%
	Cis nonProlines:	2 / 310	0.65%	Goal: <0.05%
Low-resolution Criteria	CaBLAM outliers	4	1.2%	Goal: <1.0%
	CA Geometry outliers	3	0.91%	Goal: <0.5%
Additional validations	Chiral volume outliers	0/395		

Figure 5.6 MolProbity analysis of Robetta HBHA

Protein Geometry	Poor rotamers	0	0.00%	Goal: <0.3%
	Favored rotamers	252	98.82%	Goal: >98%
	Ramachandran outliers	3	0.93%	Goal: <0.05%
	Ramachandran favored	313	96.90%	Goal: >98%
	Rama distribution Z-score	1.36 ± 0.45		Goal: abs(Z score) < 2
	Cβ deviations >0.25Å	0	0.00%	Goal: 0
	Bad bonds:	2 / 2577	0.08%	Goal: 0%
	Bad angles:	3 / 3488	0.09%	Goal: <0.1%
Peptide Omegas	Cis Prolines:	0 / 22	0.00%	Expected: ≤1 per chain, or ≤5%
	Cis nonProlines:	1 / 302	0.33%	Goal: <0.05%
Low-resolution Criteria	CaBLAM outliers	9	2.8%	Goal: <1.0%
	CA Geometry outliers	4	1.25%	Goal: <0.5%
Additional validations	Chiral volume outliers	0/383		

Figure 5.7 MolProbity analysis of Robetta HBHA Conserved

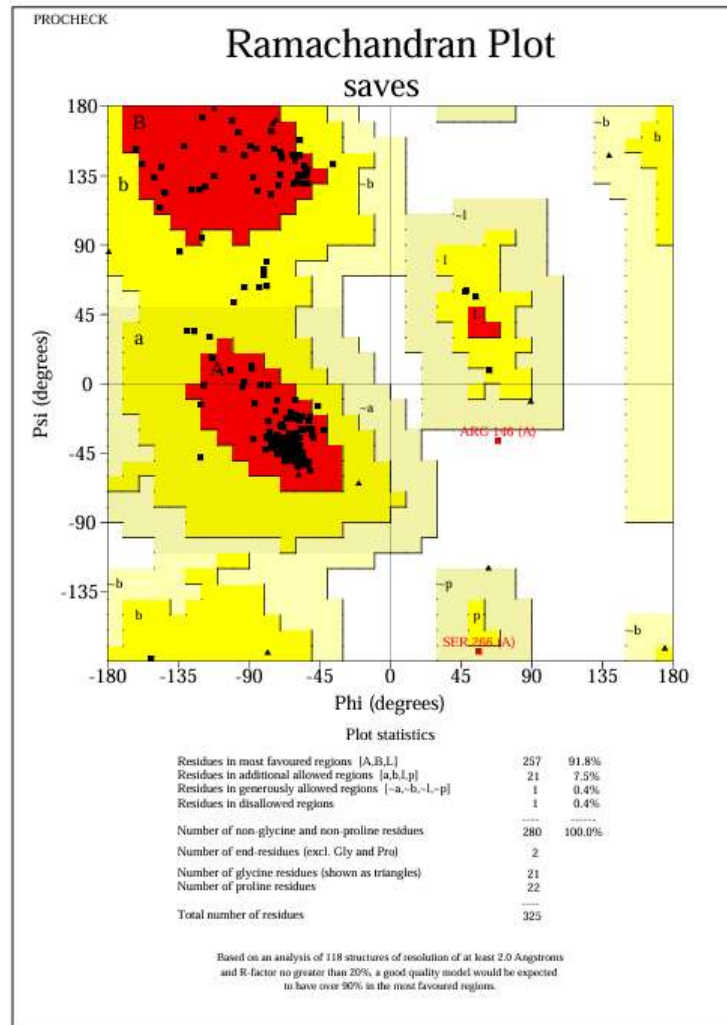


Figure 5.8 *In-silico* validation of the vaccine construct (Vc4) using Ramachandran Plot showing more than 90 percent of the residues in favourable region.

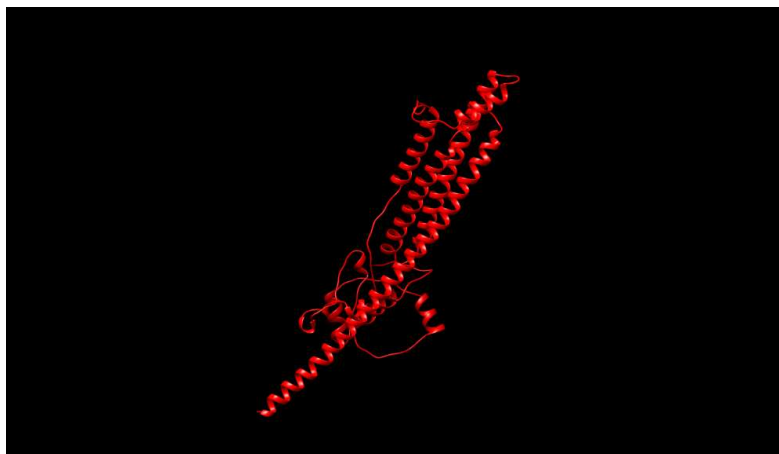


Figure 5.9 3D Structure of Vc4 modelled with the help of Robetta

5.6 Molecular Docking

To evaluate the immunological relevance of the final multi-epitope vaccine construct, molecular docking was performed with key immune recognition molecules: selected MHC class I and II alleles (HLA-B*07:02, HLA-DRB1*15:01, HLA-DRB3*01:01, HLA-B*40:01) and Toll-like receptor 4 (TLR4-3fxi) with V_C 4 to evaluate its binding affinity with each target. The lowest binding energies were seen with HLA-B*07:02 (-902.22 kJ/mol), HLA-DRB1*15:01 (-844.95 kJ/mol), HLA-DRB3*01:01(-613.83 kJ/mol), TLR4- 3fxi (-572.98 kJ/mol), HLA-B*40:01(-423.60 kJ/mol) refer to Figure 5.11 to 5.15.

5.6 Analysis of Docked complexes

These docking simulations help predict the vaccine's binding potential and immune system activation capability. Docked complexes were analysed using DIMPLOT module of LigPlot.

5.6.1 Vaccine Interaction with HLA Alleles

- **HLA-B*07:02-Vaccine Complex(ChainA)**

The analysis highlighted the formation of two hydrogen bonds, specifically between Arg264(A) and Glu229(B) (H-bond length: 3.37 Å) and Glu53(A) and Ser2(B) (H-bond length: 2.89 Å), indicating stable and favourable binding interactions. In addition, numerous hydrophobic contacts were also observed involving residues such as Ala267(A), Gly268(A), Leu275(A), Val247(B), Val248(B), Ile194(B), and Gln226(B), as shown in figure 5.16.

- **HLA-B*40:01(ChainB)-Vaccine Complex(ChainA)**

2D interactions analysis revealed three key hydrogen bonds formed between Trp308(A)–Glu44(B) (with bond length: 3.21 Å), Arg279(A)–Lys41(B) (with bond length: 2.77 Å), and Arg277(A)–Glu77(B) (with bond length: 3.51 Å), indicating a strong and stable association. Additionally, hydrophobic contacts were observed involving residues His321(A), His323(A), Val278(A), Phe304(A) and His325(A) of vaccine construct and residues Lys75(B), Asn42(B) of HLA-B*40:01 further stabilizing the complex. These interactions support the vaccine's potential for effective HLA binding and T-cell activation, as shown in Fig. 5.20.

- **HLA-DRB3*01:01(ChainB)-Vaccine Complex(ChainA)**

Interaction analysis revealed only one hydrogen bond between Arg239(A) and Asp27(B) (with bond length: 2.93 Å). Additionally, several hydrophobic contacts were observed involving residues Thr242(A), Pro229(A), Lys28(A), Leu30(A), Asn232(A),

Lys16(A), Asn34(A) from the vaccine, and Phe26(B), Glu4(B), Glu3(B), Pro96(B), Ser95(B), Phe51(B), His5(B) from HLA-DB3, supporting complex stability as shown in Fig.5.18.

- **HLA-DRB1*15:01 (ChainB)-Vaccine Complex(ChainA)**

The interaction of the vaccine construct with HLA-DRB1*15:01 involves six hydrogen bonds, enhancing stability of complex. These occur between Arg264(A)–Glu98(B) (3.38 Å), Arg220(A)–Asp181(B) (2.87 Å), Arg49(A)–Asp171(B) (3.25 Å), Glu42(A)–Leu175(B) (3.08 Å), Asn39(A)–Asp162(B) (2.20 Å), and Glu35(A)–Gly125(B) (2.73 Å). Additionally, hydrophobic interactions involve residues such as Arg43(A), Glu46(A), Thr32(A) of vaccine construct and residues Val97(B), Pro96(B), Leu174(B), Val160(B) of HLA-DRB1*15:01 supporting strong binding and compatibility, as shown in Fig.5.17.

5.6.2 Vaccine Interaction with TLR4

- **TLR4 (ChainB)-Vaccine Complex(ChainA)**

Molecular interactions showed that Arg264 (A) formed two hydrogen bonds with and Asp84(B) and Asp60(B) via its terminal NH1 group: one with the OD2 atom of Asp84 at a distance of 2.33 Å and another with the OD1 atom of Asp60 at 2.64 Å. The side chains of Gln39, Ser62, and Ser86 participate in non-covalent interactions, likely hydrophobic interactions, as indicated by red interaction arcs. These interactions suggest a stabilizing role for Arg264 in the structural integrity with binding energy of -572.98 kJ/mol as shown to Fig. 5.19.

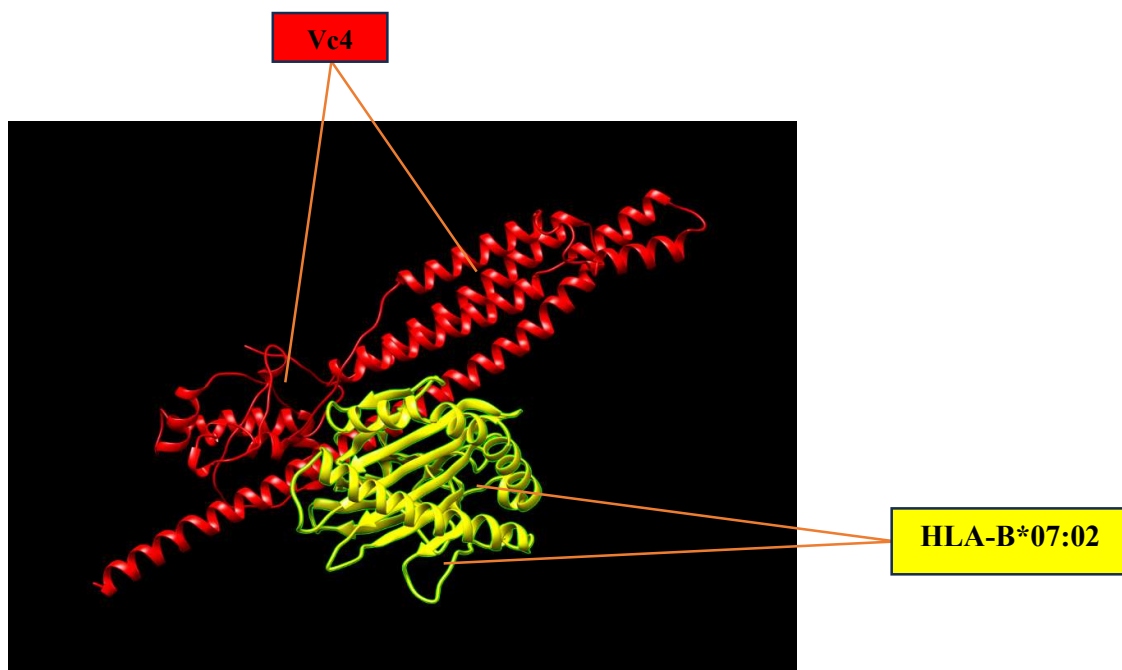


Figure 5.10 Visualisation of vaccine construct (Red) docked with HLA-B*07:02 (Yellow) in ribbon format in Chimera.

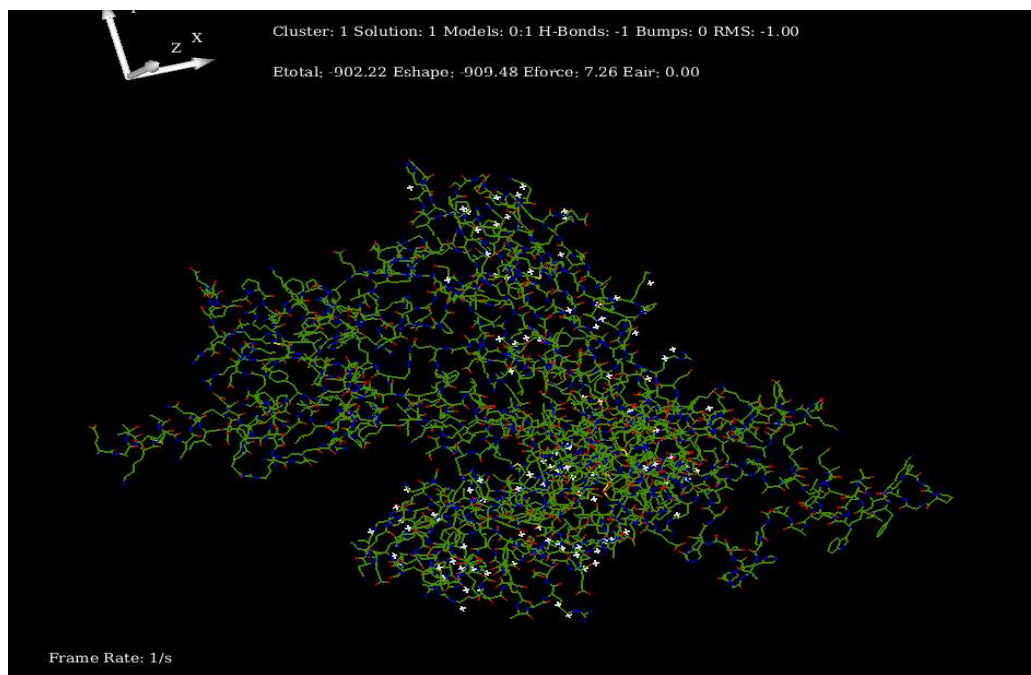


Figure 5.11 Vaccine construct docked with HLA-B*07:02 which was showing highest binding affinity (HEX docking).

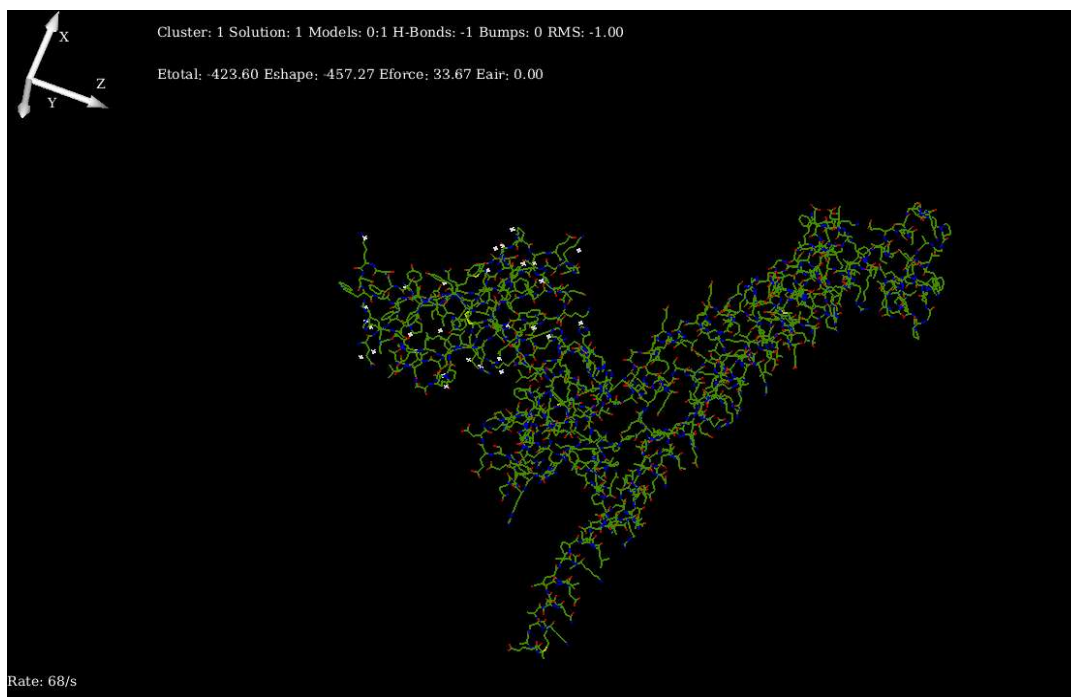


Figure 5.12 Vaccine construct docked with HLA-B*40:01 which was showing with lowest affinity (HEX docking).

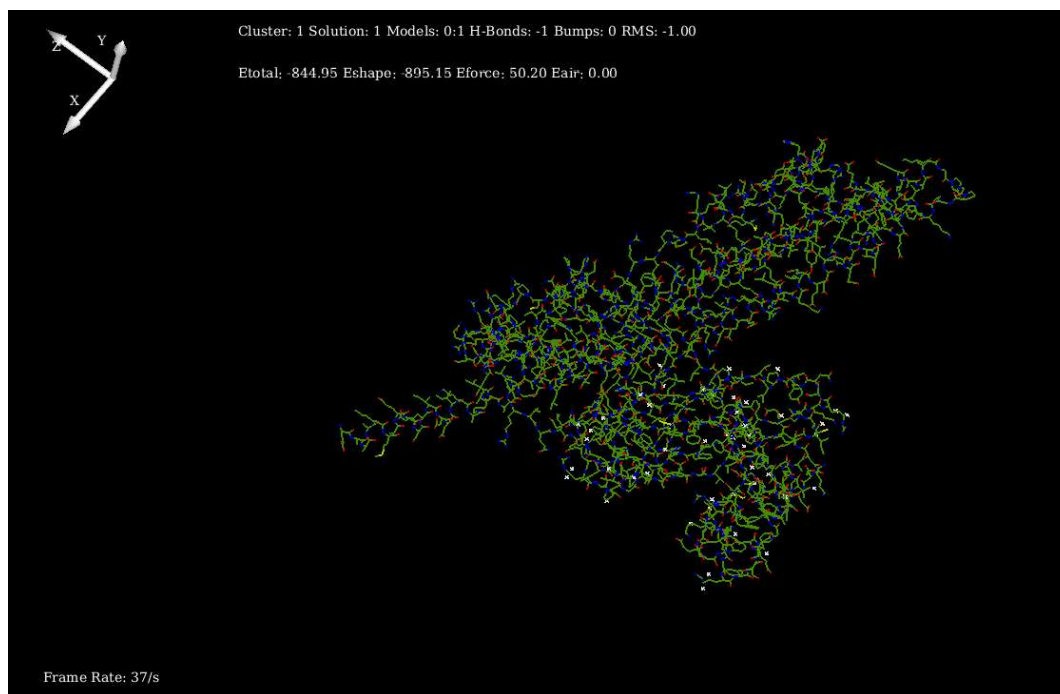


Figure 5.13 Vaccine construct docked with HLA-DRB1*15:01 which was showing second highest affinity (HEX docking).

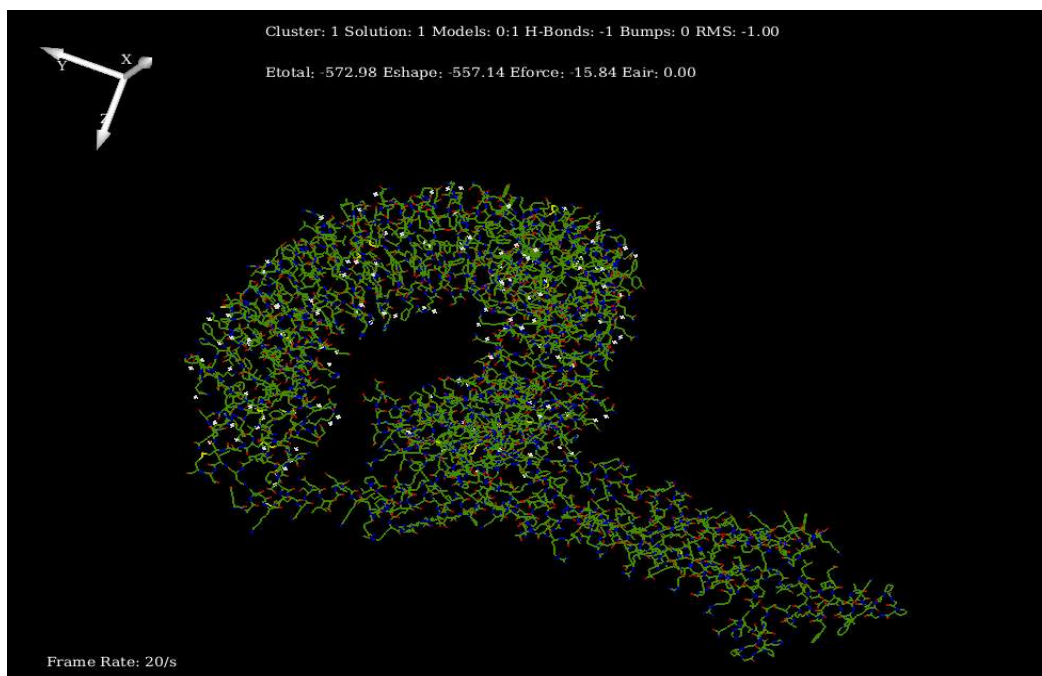


Figure 5.14 Vaccine construct docked with TLR4 which was showing moderate affinity (HEX docking).

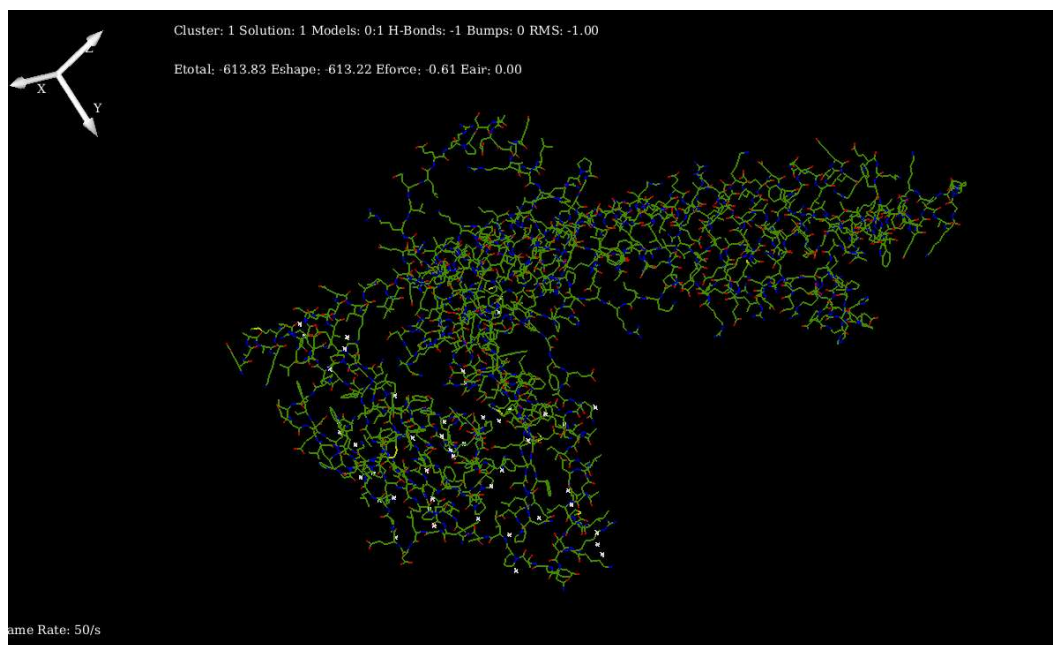


Figure 5.15 Vaccine construct docked with HLA-DRB3*01:01 which was showing moderate affinity (HEX docking).

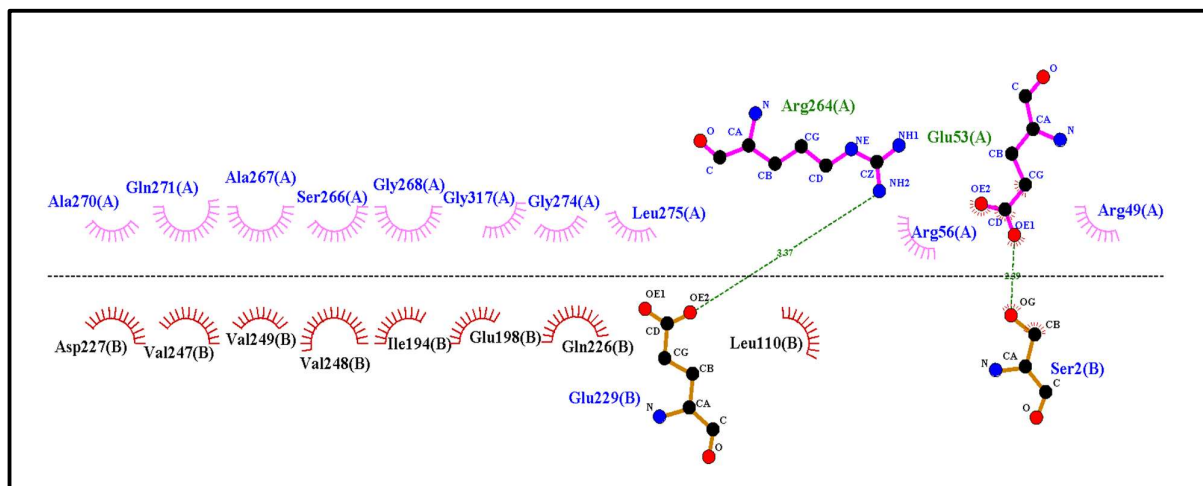


Figure 5.16 2D interaction between Vc4 and HLA-B*07:02. Hydrogen bonds are shown with Green dashed line (-----) Hydrophobic interactions shown in Pink (for chain A) and Red (for chain B).

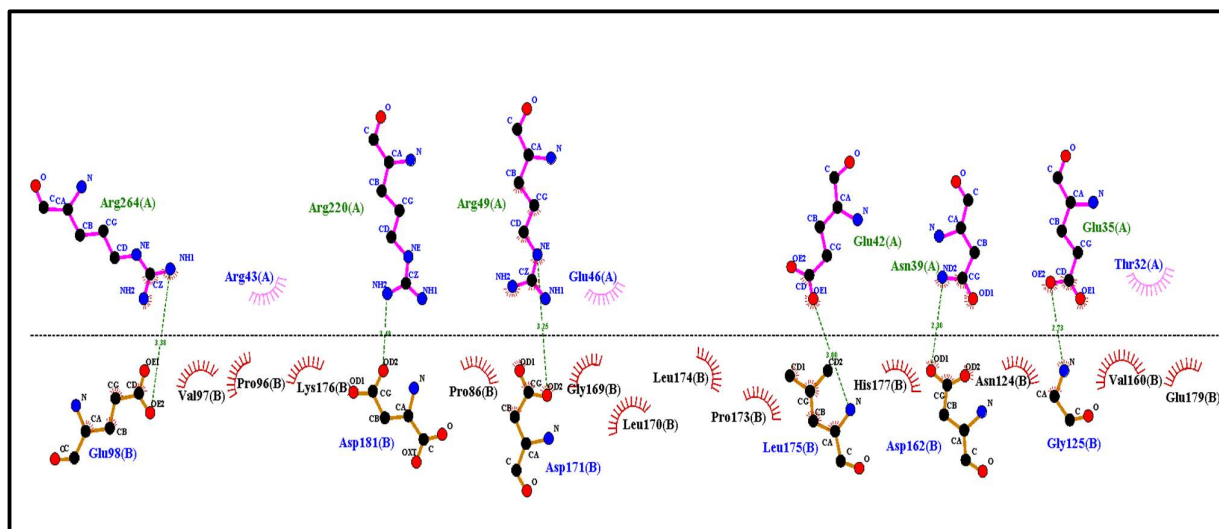


Figure 5.17 2D interaction between Vc4 and HLA-DRB1*15:01. Hydrogen bonds are shown with Green dashed line (-----) Hydrophobic interactions shown in Pink (for chain A) and Red (for chain B).

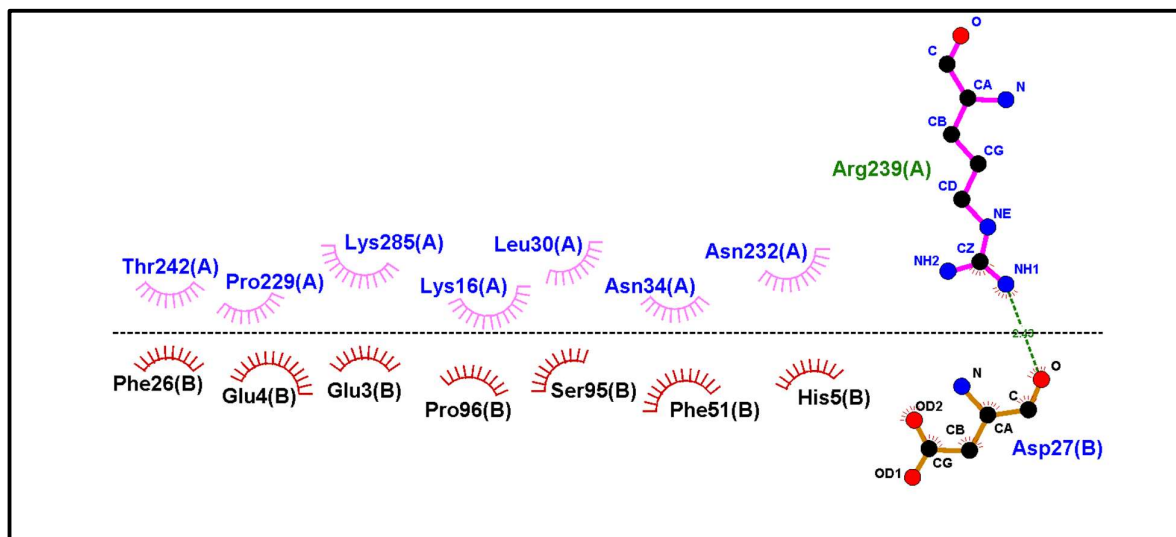


Figure 5.18 2D interaction between Vc4 and HLA-DRB3*01:01. Hydrogen bonds are shown with Green dashed line (---) Hydrophobic interactions shown in Pink (for chain A) and Red (for chain B).

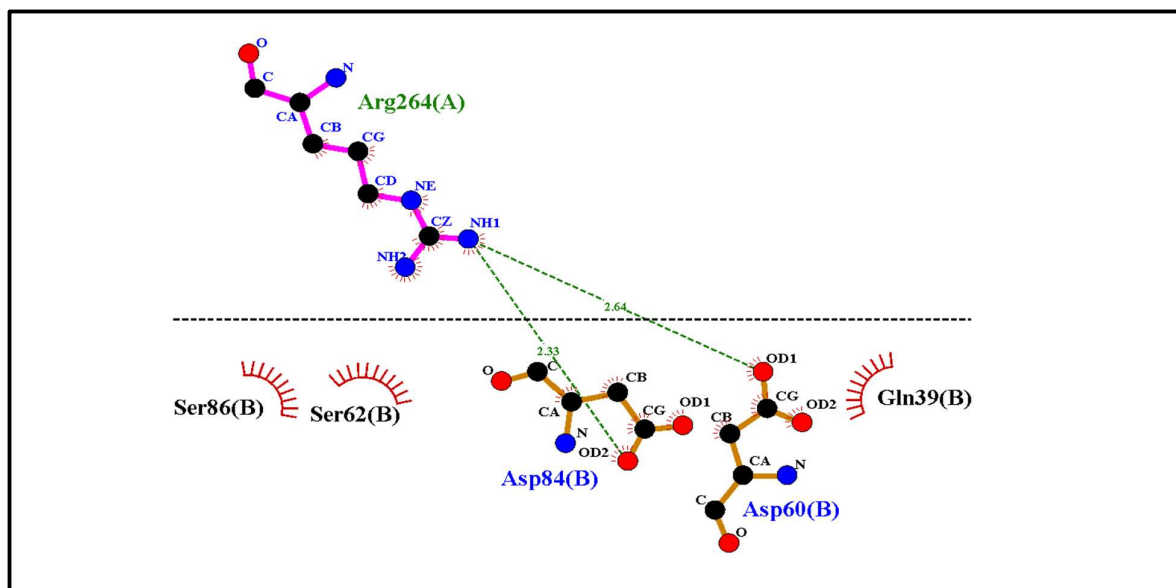


Figure 5.19 2D interaction between Vc4 and TLR4. Hydrogen bonds are shown with Green dashed line (---) Hydrophobic interactions shown in Pink (for chain A) and Red (for chain B).

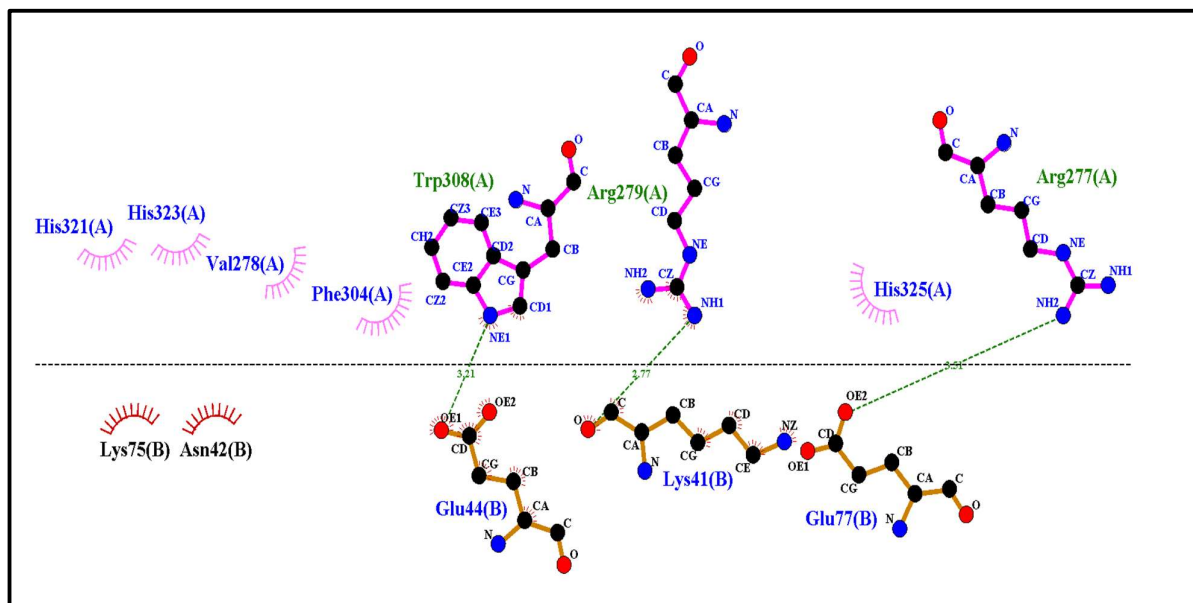


Figure 5.20 2D interaction between Vc4 and HLA-B*40:01. Hydrogen bonds are shown with Green dashed line (---) Hydrophobic interactions shown in Pink (for chain A) and Red (for chain B).

CHAPTER 6: CONCLUSION

HCMV is one of the major global health threats, and it is present in a significant portion of the world's population. The results of seroprevalence studies show that an average of 50% to 90% of the global adult population is infected with HCMV over time. There was a dire need for an effective and available vaccine, given the wide range of people's exposure to HCMV and the disability it can cause to their immune systems. Despite extensive research, no licensed and safe HCMV vaccine exists to date. Conventional strategies—including live-attenuated, subunit, and vector-based vaccines which face major limitations. Therefore, this study presents a systematic and integrative *in-silico* approach for the rational design of a multi-epitope vaccine candidate targeting Human Cytomegalovirus (HCMV), a globally prevalent pathogen associated with severe clinical outcomes, particularly in neonates and immunocompromised individuals. The limitations of conventional vaccine strategies largely stemming from the virus's genomic complexity, latency, and immune evasion mechanisms necessitate the exploration of computational methodologies to accelerate and enhance vaccine development. To overcome these limitations a reverse vaccinology pipeline was used where antigenic proteins from the HCMV AD169 strain was identified and comprehensively analyzed for epitope prediction. High-affinity cytotoxic T lymphocyte (CTL), helper T lymphocyte (HTL), and B-cell epitopes were screened using established bioinformatics tools.

With antigenicity and allergenicity prediction we obtained HCMVTRL1 = IRL1 as a potential candidate for multi-epitope vaccine construction. Potential epitopes were predicted and re-evaluated for their immunogenicity, allergenicity, toxicity and antigenicity. Obtained epitopes were utilised for multi-epitope construction in various combinations. The final construct showed lowest binding energies with HLA-B*07:02 (-902.22 kJ/mol), HLA-DRB1*15:01 (-844.95 kJ/mol), HLA-DRB3*01:01(-613.83 kJ/mol), TLR4 (-572.98 kJ/mol), and HLA-B*40:01(-423.60 kJ/mol). The interactions between vaccine construct and target proteins also support the stable complex formation. So, this construct can be a potential vaccine which needs further *in-silico* and *in-vivo* studies to conform these analyses.

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